SLEC7000

This manual describes the correct use of the SLEC7000

SELOON Elevator Co.,Ltd

Preface

Thank you for purchasing the SLEC7000 series integrated elevator controller. The SLEC7000 is a new-generation integrated elevator controller independently developed and manufactured by Seloon Elevator Co.,Ltd. The SLEC7000 has the following advantages:

- It supports high-performance vector control. It can drive both AC asynchronous motor and permanent magnetic synchronous motor (PMSM), and implement switchover between the two types of motors easily by modifying only one parameter.
- It supports open-loop low-speed running, group control on a maximum of eight elevators (no additional external device required), and CANbus and Modbus communication protocols for remote monitoring, which reduces the required quantity of traveling cables
- It supports a maximum of 56 floors and is widely applied to elevators used in the villa and freight elevators.

This manual describes the correct use of the SLEC7000, including product information, installation, common parameter setting, and commissioning. Read and understand the manual before using the product, and keep it carefully for reference to future reference.

Notes
● The drawings in the manual are sometimes shown without covers or protective guards. Remember to install the covers or protective guards as specified first, and then perform operations in accordance with the instructions. ● The drawings in the manual are shown for description only and may not match the product you purchased. ● The instructions are subject to change, without notice, due to product upgrade, specification modification as well as efforts to increase the accuracy and convenience of the manual. ● Contact our agents or customer service center if you need a new user manual or have problems during the use.

1.1Terminal Description

The following figure shows the terminal arrangement and description of the SLEC7000.

Figure 1-3 Terminal arrangement and description of the SLEC7000

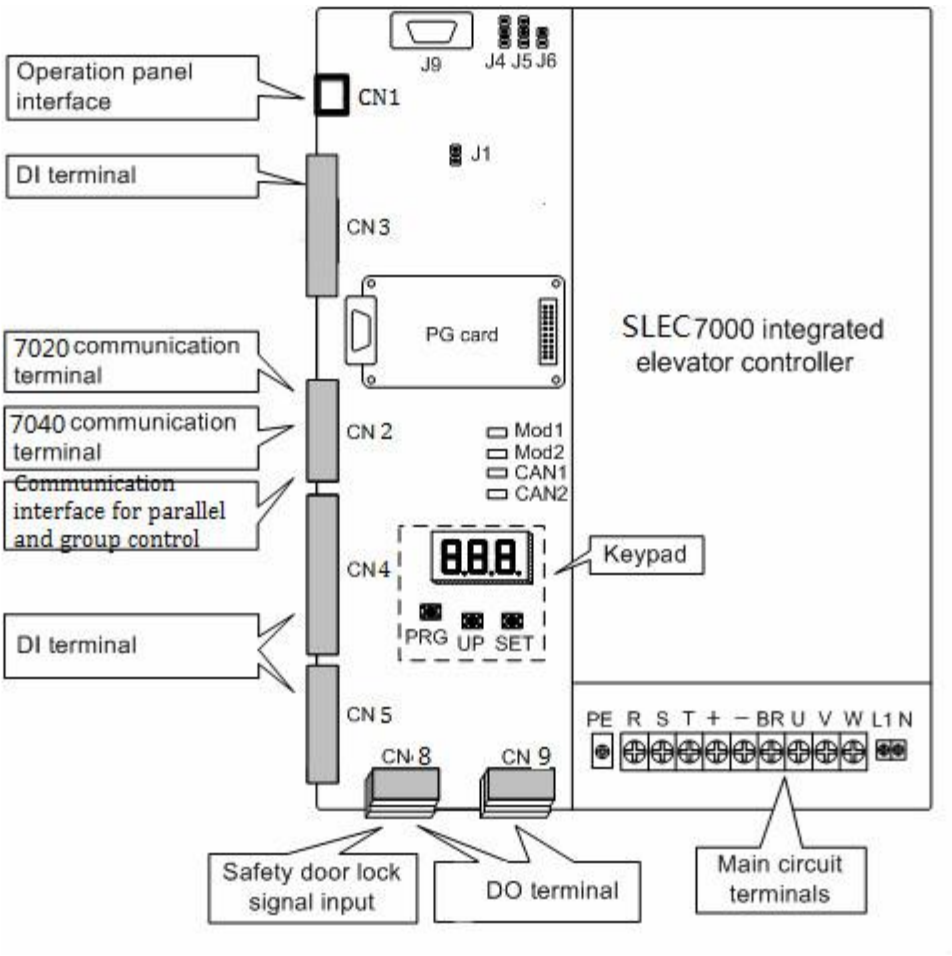
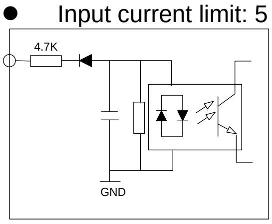


Table 1-4 Terminal functions

Mark	Code	**Terminal Name**	**Function Description**
CN3 CN4 CN5 CN6 CN7	X1 to X28	DI	● Input voltage range: 10–30 VDC ● Input impedance: 4.7 kΩ ● Optocoupler isolation ● Input current limit: 5 mA  Functions set in F5-01 to F5-28 Slow-down switches connected to terminals among X1 to X6 (recommended)
CN2	+24V/MCM	24 V power supply	24 VDC power supply for the MCB, used for input, output and communication circuits.
	Mod1+/Mod1-	Modbus communication terminal	Hall call board (SSL7040) serial communication terminal
	CAN1+/CAN1-	CANbus communication terminal	Car top board (SSL7020) CAN communication terminal
CN8	Y1/Y2/Y3/YM	Relay output	Normally-open (NO), maximum current and voltage rating: 5 A, 250 VAC

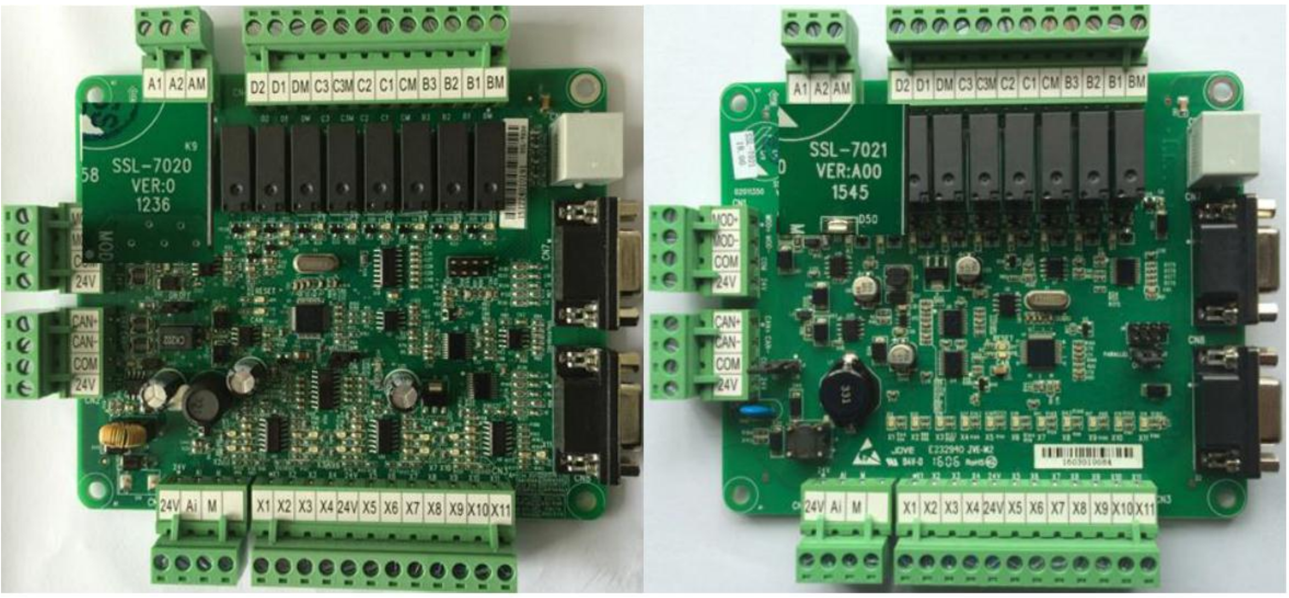
Mark	Code	Terminal Name	Function Description
			Function set in F5-32 to F5-34
	X29, X30, X31 to XCOM	DI	Input voltage range: 95–125 VAC Safety circuit and door lock circuit, function set in F5-29 to F5-31
CN9	Y7-M7~Y9-M9 Y4、Y5、Y6、 YM2	Relay output	Normally-open (NO), maximum current and voltage rating: 5 A, 250 VAC Function set in F5-36 to F5-40
CN2	Mod2+/Mod2-	Modbus communication terminal	Used for remote monitor or standby SSL7040 communication
	CAN2+/CAN2-	CANbus communication terminal	Used for CAN communication for parallel or group control

Table 1-5 **Description of indicators on the MCB**

Mark	Terminal Name	Function Description
HOP	Standby communication indicator	This indicator blinks (green) when the communication is normal.
COP	Standby communication indicator	This indicator is steady on (green) when the communication for parallel/group control is normal, and blinks when the running in parallel/group mode is normal.
OK	SSL7040 communication indicator	This indicator is on (green) when the communication between the SSL-7000-A1 and the SSL7040 is normal.
ER	SSL7020 communication indicator	This indicator blinks (green) when the communication between the SSL-7000-A1 and the SSL7020 is normal.
X1 to X31	Input indicator	This indicator is on when the external input is active.
Y1 to Y9	Output indicator	This indicator is on when the system output is active.

CTB board

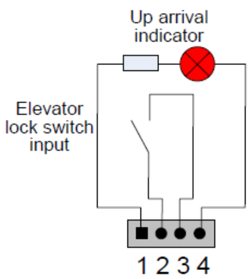
SSL7020/SSL7021

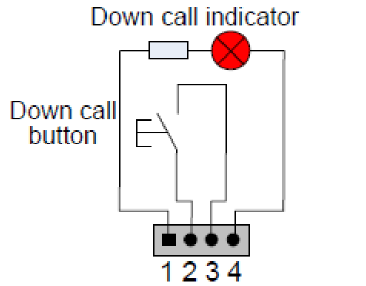
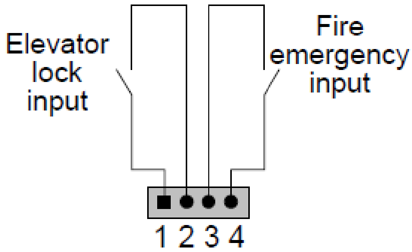
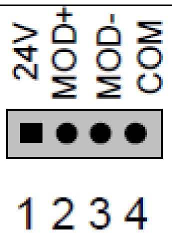


Mark		Terminal name	Function description
CN2	+24V/COM	External 24V DC power supply	24VDC power supply for SSL7020
	CAN+/CAN-	CAN bus communication interface	Connecting the SLEC7000 for CAN bus communication
CN1	+24V/COM	DC24V power supply	24VDC power supply for SSL7020
	MOD+/MOD-	Mod buss communication	Connecting the SSL7020 for mod bus communication
CN6	Ai-M	Load cell signal input	0-10 VDC
CN3	X1	Light curtain 1	DI terminal 1.photocoupler isolation, unipolarity input 2.input impedance:3.3kΩ
	X2	Light curtain 2	
	X3	Door open limit 1	
	X4	Door open limit 2	
	24V	24 VDC power supply	DI common terminal
	X5	Door close limit 1	DI terminal 1.photocoupler isolation, unipolarity input 2.input impedance:3.3kΩ
	X6	Door close limit 2	
	X7	Full load signal input	
	X8	Over load signal input	
	X9	No load signal input	
	X10	Safety edge 1	
	X11	Safety edge 2	
CN4	BM	Door 1 common port	Relay output terminal contact drive capacity:30 VDC,1A
	B1	Door open signal 1	
	B2	Door close signal 1	
	B3	Forced door close 1	
	CM	Door 2 common port	
	C1	Door open signal 2	
	C2	Door close signal 2	
	C3	Forced door close 2	
	DM	Arrival common port	
	D1	Up arrival signal	
	D2	Down arrival signal	
	A1-1M(NC contact) A2-AM(NC contact)	Car fan and lamp control	Relay output terminal contact drive capacity:250VAC,3A or 30 VDC,1A
CN7/CN8		DB9-pin interface for communication with the SSL7030	Connecting the SSL7030
	CN10	RJ45 interface	Connecting the external operation panel
	CAN	CANbus communication indicator	This indicator blinks when communication between ssl7020 and the ssl7030 is normal, and is steady on when a communication fault occurs
	RESET	CANbus communication fault indicator	

	X1~X11	DI indicator	This indicator is on(green) when the external input is active
	A1~D2	Relay output indicator	This indicator is on(green) when the system output is active
	J9	reserved	It is factory reserved. do not short it randomly. otherwise, the controller may not be used properly

1.2 SSL7040

Remark	Terminal Name	Function Description	Terminal Arrangement
J2	Set the physical address	Set the floor address: Short J2, press on up、down call floor to set address, remove the jumper, address storage. (Range 0 to 56)	
JP1/JP2	UP1/UP2	Interface for the elevator lock switch and up arrival indicator Pins 2 and 3 are for switch input. Pins 1 and 4 are output of the up arrival indicator	

JP3/JP4	DOWN1/DOWN2	Interface for the elevator lock switch and down arrival indicator Pins 2 and 3 are for switch input. Pins 1 and 4 are output of the down arrival indicator	
JP5	XF/ST	Interface for the fire emergency and elevator lock switches Pins 1 and 2 are for fire emergency input. Pins 3 and 4 are for elevator lock input.	
CN1	MOD bus communication	MOD bus communication and power supply	

1.3 SSL7030/SSL7031(Instruction plate inside the car)


SSL7030

Terminal function

NO.	Interface	2、3 pins	1、4 pins
1	JP1	Button input of floors 1	Button output of floors 1
2	JP2	Button input of floors 2	Button output of floors 2
3	JP3	Button input of floors 3	Button output of floors 3
4	JP4	Button input of floors 4	Button output of floors 4
5	JP5	Button input of floors 5	Button output of floors 5

NO.	Interface	2、3 pins	1、4 pins
6	JP6	Button input of floors 6	Button output of floors 6
7	JP7	Button input of floors 7	Button output of floors 7
8	JP8	Button input of floors 8	Button output of floors 8
9	JP9	Button input of floors 9	Button output of floors 9
10	JP10	Button input of floors 10	Button output of floors 10
11	JP11	Button input of floors 11	Button output of floors 11
12	JP12	Button input of floors 12	Button output of floors 12
13	JP13	Button input of floors 13	Button output of floors 13
14	JP14	Button input of floors 14	Button output of floors 14
15	JP15	Button input of floors 15	Button output of floors 15
16	JP16	Button input of floors 16	Button output of floors 16
17	JP17	Button input of door open	Display output of door open
18	JP18	Button input of door close	Display output of door close
19	JP19	Button input door open delay	Display output of door open delay
20	JP20	Direct Input	Stop car output in non-door machine
21	JP21	Driver input	Reserved
22	JP22	Commutation input	Reserved
23	JP23	Run independently input	Reserved
24	JP24	Firefighters input	Reserved
Note:PCB board with white dots mark or welding pin of square is 1 pin.			

Chapter 2 Use Description

2.1 Use of the Onboard Keypad

The onboard keypad consists of three 7-segment LEDs and three buttons. You can view information about the controller and enter simple commands on the keypad.

The following figure shows the appearance of the keypad.

Figure 2-1 Appearance of the keypad

As shown in the preceding figure, the three buttons are PRG, UP, and SET. The functions of the three buttons are as follows:

- PRG: Press this button in any state to display the current function group number.

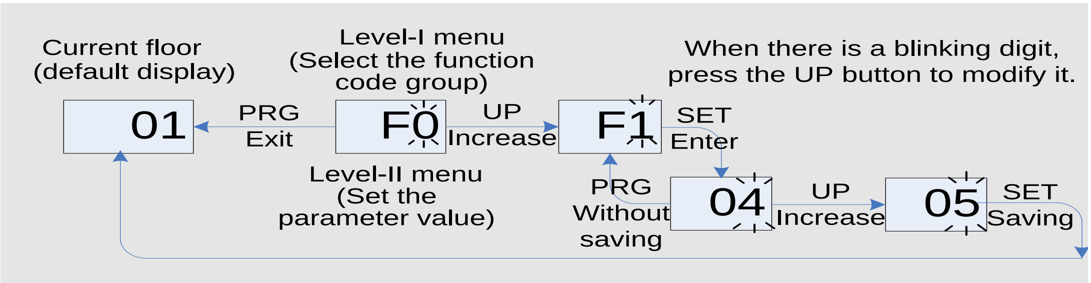
  You can press the UP button to change the function group number.
- UP: Press this button to increase the function group number.

  Currently, the SSL-7000-A1 defines a total of 11 function code groups, namely, F0 to F9, and FA. You can press the UP button to display them in turn. In addition, in special function code group menu, you can input simple references by using the UP button.
- SET: In the function code group menu, press this button to enter the menu of the function code group.

  In special function code group menu, after you input a simple reference and press this button to save the setting, the display will return to the F0 menu by default.

  The following figure shows the setting of increasing the called floor to 5.

  Figure 2-2 Setting the called floor



The function code groups displayed on the keypad are described as follows:

1. F0: display of floor and running direction

   The F0 menu is displayed on the keypad by default upon power-on. The first LED indicates the running direction, while the last two LEDs indicate the current floor number of the elevator.
   When the elevator stops, the first LED has no display. When the elevator runs, the first LED blinks to indicate the running direction.
   When a system fault occurs, the 7-segment LEDs automatically display the fault code and blink. If the fault is reset automatically, the F0 menu is displayed.
2. F1: command input of the running floor

After you enter the F1 menu, the 7-segment LEDs display the bottom floor (F6-01). You can press the UP button to set the destination floor within the range of lowest to top and then press the SET button to save the setting. The elevator runs to the destination floor, and the display switches over to the F0 menu at the same time.

3. F2: fault reset and fault code display

   After you enter the F2 menu, the 7-segment LEDs display "0". You can press the UP button to change the setting to 1 or 2.
   Display "1": If you select this value and press the SET button, the system fault is reset. Then, the display automatically switches over to the F0 menu.
   Display "2": If you select this value and press the SET button, the 7-segment LEDs display the 20 fault codes and occurrence time circularly. You can press the PRG button to exit.

4. F3: time display

   After you enter the F3 menu, the 7-segment LEDs display the current system time circularly.

5. F4: contract number display

   After you enter the F4 menu, the 7-segment LEDs display the user's contract number.

6. F5: door open/close control

   After you enter the F6 menu, the 7-segment LEDs display "1-1", and the UP and SET buttons respectively stand for the door open button and door close button. You can press the PRG button to exit.

7. F6: reserved

8. F7: shaft auto-tuning command input

   After you enter the F7 menu, the 7-segment LEDs display "0". You can select 0 or 1 here, where "1" indicates the shaft auto-tuning command available.
   After you select "1" and press the SET button, shaft auto-tuning is implemented if the conditions are met. Meanwhile, the display switches over to the F0 menu. After shaft auto-tuning is complete, F7 is back to "0" automatically. If shaft auto-tuning conditions are not met, fault code "E35" is displayed.

9. F8: test function

   After you enter the F8 menu, the 7-segment LEDs display "0". The setting of F8 is described as follows:

   • 00: No function

   • 01: Hall call forbidden

   • 02: Door open forbidden

   • 03: Overload forbidden

   • 04: Limit switches disabled

   After the setting is complete, press the SET button. Then the 7-segment LEDs display "E88" and blink, prompting that the elevator is being tested. When you press PRG to exit, F8 is back to 0 automatically.

10. F9: reserved

11. FA: auto-tuning

   After you enter the FA menu, the 7-segment LEDs display "0". The setting range of FA is 0–3, as follows:
   • 00: No function
   • 01: With-load auto-tuning

   • 02: No-load auto-tuning

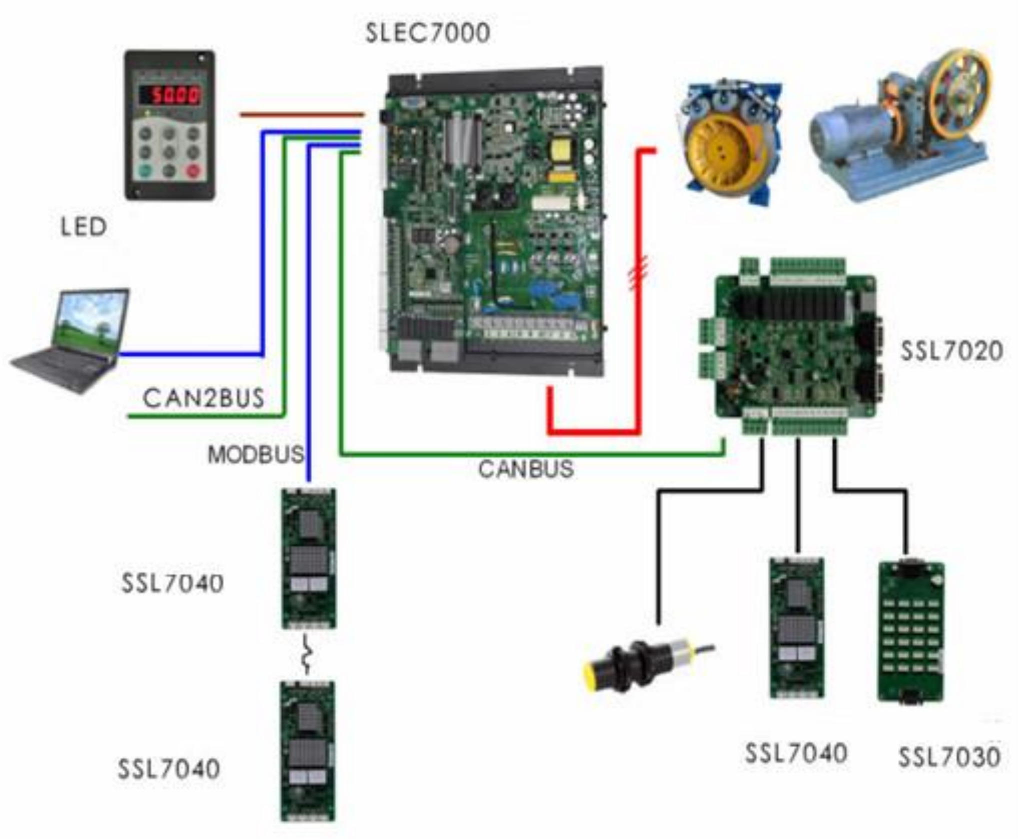
   • 03: Synchronous motor auto-tuning

   After the setting is complete, press the SET button. Then the 7-segment LEDs display "TUNE", and the elevator enters the auto-tuning state. After confirming that the elevator meets the safe running conditions, press the SET button again to start auto-tuning.
   After auto-tuning is complete, the 7-segment LEDs display the current angle for 2s, and then switch over to the F0 menu.

You can press the PRG button to exit the auto-tuning state.

2.2 Electrical Wiring Diagram of the SLEC7000 Control System

Figure 2-3 Electrical wiring diagram of the SLEC7000 control system



2.3 Installation of Shaft Position Signals

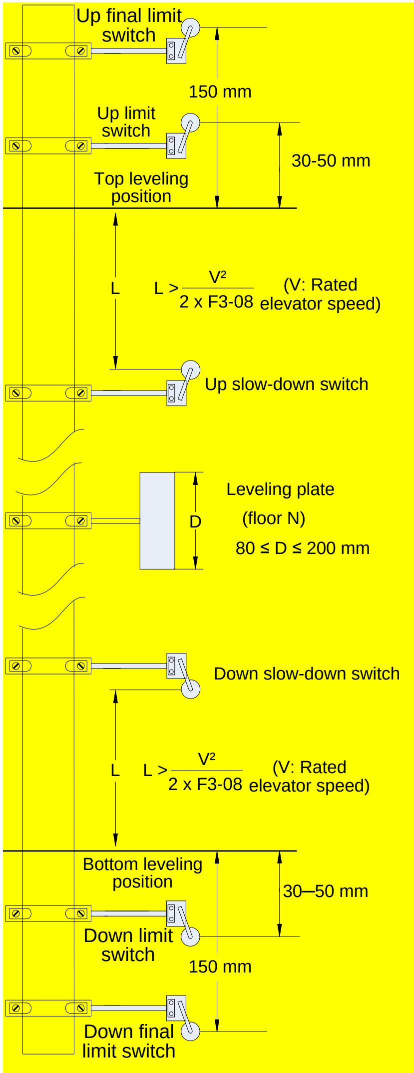
In elevator control, to implement landing accurately and running safely, the car position needs to be identified based on shaft position signals.

These shaft position signals include the leveling switches, up/down slow-down switches, up/down limit switches, and up/down final limit switches.

These shaft position signals are directly transmitted by the shaft cables to the SSL-7000-A1 of the controller. For the electrical wiring method, refer to Figure 2-3 on the last page of this chapter.

The following figure shows the arrangement of shaft position signals in the shaft.

Figure 2-4 Arrangement of shaft position signals



2.3.1 Installation of Limit Switches and Final Limit Switches

The up limit switch and down limit switch protect the elevator from over travel top/bottom terminal when the elevator does not stop at the leveling position of the terminal floor.

- The up limit switch needs to be installed 30−50 mm away from the top leveling position. The limit switch acts when the car continues to run upward 30−50 mm from the top leveling position.
- The down limit switch needs to be installed 30−50 mm away from the bottom leveling position. The limit switch acts when the car continues to run downward 30−50 mm from the bottom leveling position.

The final limit switch is to protect the elevator from over travel top/bottom terminal when the elevator does not stop completely upon passing the up/down limit switch.

- The up final limit switch is mounted above the up limit switch. It is usually 150 mm away from the top leveling position.
- The down final limit switch is mounted below the down limit switch. It is usually 150 mm away from the bottom leveling position.

2.3.2 Installation of Slow-Down Switches

The slow-down switch is one of the key protective components of the SLEC7000, protecting the elevator from over travel top terminal or over travel bottom terminal at maximum speed when the elevator position becomes abnormal.

The slow-down distance L indicates the distance from the slow-down switch to the leveling plate at the terminal floor. The calculating formula is as follows:

$$L > \frac{V^2}{2 \times \text{F3-08}}$$

In the formula, "L" indicates the slow-down distance, "V" indicates the F0-04 (Rated elevator speed), and "F3-08" indicates the special deceleration rate.

The default value of F3-08 (Special deceleration rate) is 0.9 m/s^2. The slow-down distances calculated based on different rated elevator speeds are listed in the following table:

Table 2-1 Slow-down distances based on different rated elevator speeds

Rated Elevator Speed	V ≤ 1.5 m/s	1.5 m/s < V ≤ 2.4 m/s	2.4 m/s < V ≤ 3.7 m/s
Distance of slow-down 1	1.3 m to H/2	1.3 m	1.3 m
Distance of slow-down 2	-	3.2 m	3.2 m
Distance of slow-down 3	-	-	8.0 m

Note

- H indiates the height of the landing floor. The slow-down distances above are calculated on the basis of the default values (special deceleration rate 0.9 m/s^2, and acceleration rate and deceleration rate 0.6 m/s^2).
- Increasing the acceleration rate and deceleration rate or reducing the special deceleration rate may bring safety hazard. If any change is in need, re-calculate the slow-down distance by using the above formula.

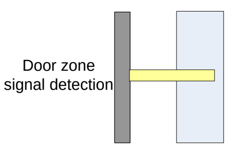
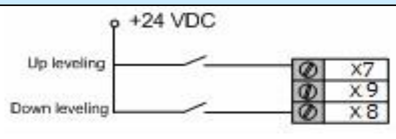
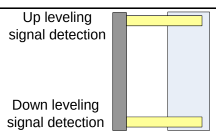
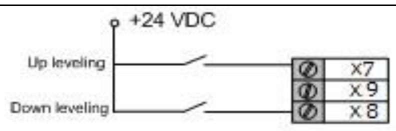
2.3.3 Installation of Leveling Signals

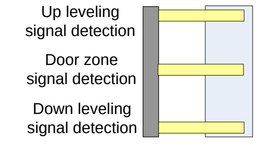
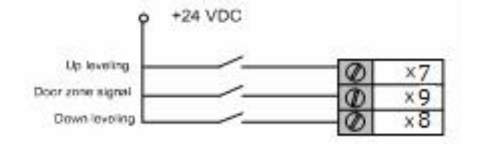
Leveling signals comprise the leveling switch and leveling plate and are directly connected to the input terminal of the controller. It is used to enable the car to land at each floor accurately.

The leveling switches are generally installed on the top of the car. The SLEC7000 system supports the installation of 1−3 leveling switches. The leveling plate is installed on the guide rail in the shaft. A leveling plate needs to be installed at each floor. Ensure that leveling plates at all floors are mounted with the same depth and verticality.

The following table describes the installation requirements of leveling switches.

Table 2-2 Installation requirements of leveling switches

Number of Leveling Switches	**Installation Method**	**Connecting to Input Terminals of Controller**	**Setting of Function Code**
1		 Short X7 and X8	**F5-07 = 001** **F5-08 = 002（NO)** **F5-09 = 000**
2			**F5-07 = 001** **F5-08 = 002（NO)** **F5-09 = 000**

3			F5-07 = 001 F5-08 = 002（NO) F5-09 = 003

Note

- When installing leveling plates, ensure that the plates at all floors are mounted with the same depth and verticality. Otherwise, the leveling accuracy will be affected. The recommended length of the plate is 80–200 mm.
- More leveling input signals need to be added if the door pre-open function is used. In this case, you need to increase the length of the plate properly. For details on the door pre-open module, contact the local agent or Monarch.

2.4 Typicl Commissioning Description

2.4.1 Commissioning at Inspection Speed

You can change the value of F1-25 only to perform switchover between synchronous motor and asynchronous motor.

Table 2-3 Motor parameters to be set for motor auto-tuning

Function Code	Parameter Name	Description
F1-25	Motor type	0: Asynchronous motor 1: Synchronous motor
FH-01	Encoder type	0: SIN/COS encoder, absolute encoder 1: UVW encoder 4: ABZ incremental encoder
FH-02	Encoder pulses per revolution	0–10000
F1-01 to F1-05	Rated motor power Rated motor voltage Rated motor current Rated motor frequency Rated motor rotational speed	These parameters are model dependent, and you need to manually input them according to the nameplate.
F0-00	Control mode	0: Sensorless vector control (SVC) 1: Closed-loop vector control (CLVC) 2: Voltage/Frequency (V/F) control
F0-01	Command source selection	0: Operation panel control 1: Distance control
F2-11	No-load-cell startup	0: Disabled 1: Enabled
F1-11	Auto-tuning mode	0: No operation 1: With-load auto-tuning 2: No-load auto-tuning 3: Parameter auto-tuning

■ **Precautions for Motor Auto-tuning**

Follow the following precautions:

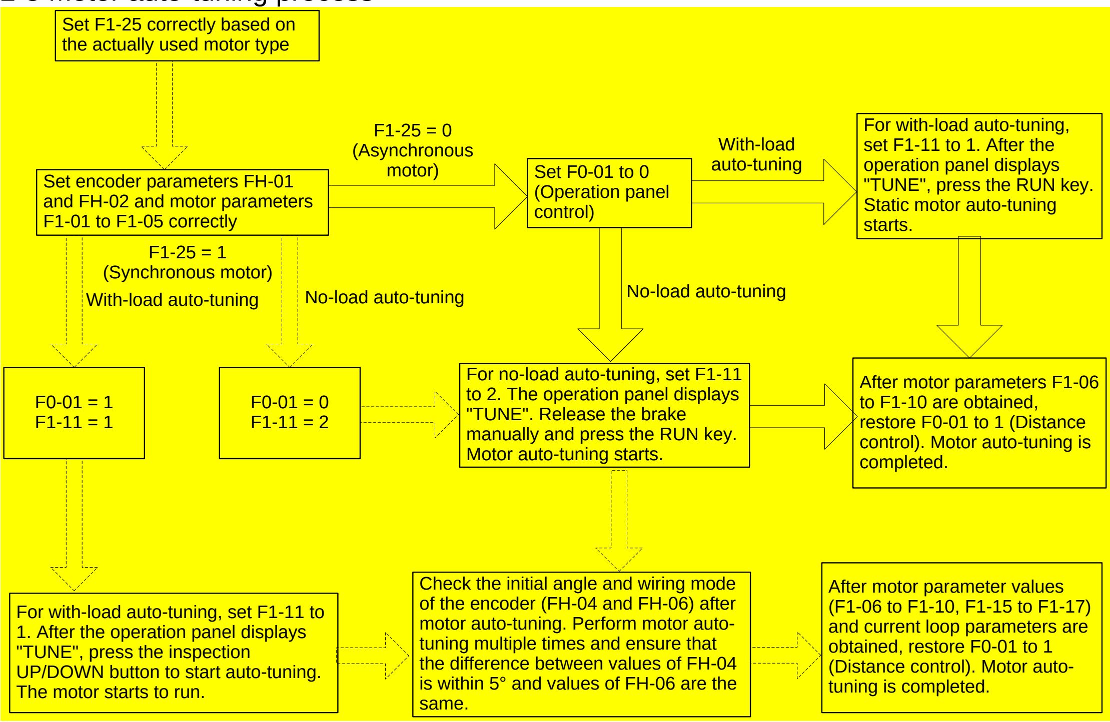
- Ensure that all wiring and installation meet the safety specifications.
- Set F1-25 (Motor type) correctly, and enter the motor parameters in group F1 (F1-01 to F1-05) correctly. Otherwise, motor auto-tunning cannot be performed.
- Set the FH-01 (Encoder type) and FH-02 (Encoder pulses per revolution) correctly. Before performing auto-tuning, ensure that F0-00 (Control mode) is 1 (Closed-loop vector control (CLVC))

and F0-01 (Command source selection) is 1 (Distance control).

- Ensure that the motor wiring is correct (UVW cables of the motor respectively connected to UVW cables of the controller) for with-load auto-tuning. If the motor wiring is incorrect, the motor may jitter or fail to run after the brake is released; in this case, you need to replace any two of the motor UVW cables.
- Reset the current fault and then start auto-tuning, because the system cannot enter the auto-tuning state ("TUNE" is not displayed) when there is a fault.
- Perform motor auto-tuning again if the phase sequence or encoder of the synchronous motor is changed.
- For the synchronous motor, perform three or more times of auto-tuning, compare the obtained values of FH-04 (Encoder initial angle). The value deviation of FH-04 shall be within ±5°, which indicates that the auto-tuning is successful.
- After the auto-tuning is completed, perform trial inspection running. Check whether the current is normal, whether the actual running direction is the same as the set direction. If the running direction is different from the set direction, change the value of F0-05.
- With-load auto-tuning is dangerous (inspection-speed running of many control cabinets is emergency electric running and the shaft safety circuit is shorted). Ensure that there is no person in the shaft in this auto-tuning mode.

The following figure shows the motor auto-tuning process.

Figure 2-5 Motor auto-tuning process



More descriptions of motor auto-tuning are as follows:

1. For synchronous motor, with-load auto-tuning learns stator resistance, shaft-D and shaft-Q inductance, current loop (including zero servo) PI parameters, and encoder initial angle; no-load auto-tuning additionally learns the encoder wiring mode.

2. For the asynchronous motor, static auto-tuning learns stator resistance, rotor resistance, and leakage inductance, and automatically calculates the mutual inductance and motor magnetizing current. No-load auto-tuning learns the mutual inductance, motor magnetizing current, and current loop parameters.

3. For synchronous motor, when F1-11 is set to 3, current loop parameters are learned when the motor is

in static mode; the brake does not release in this process. For the asynchronous motor, there is no difference regardless of whether F1-11 is 1 or 3.

4. The system learns the current loop parameters during auto-tuning by default. If the riding comfort is satisfactory with the setting of the current loop parameters, you can set Bit2 of FA-12 to 1 to disable the current loop PI self-adaptation function during re-auto-tuning.

2.4.2 Inspection at Normal Speed

After ensuring that running at inspection speed is normal, perform shaft auto-tuning, and then you can perform trial running at normal speed (the elevator satisfies the safety running requirements).

To perform shaft auto-tuning, the following conditions must be satisfied:

1. The signals of the encoder and leveling sensors (NC, NO) are correct and the shaft position switches are installed properly and act correctly.

2. When the elevator is at the bottom floor, the down slow-down 1 switch acts.

3. The elevator is in the inspection state. The control mode is distance control and CLVC (F0-00 = 1, F0-01 = 1).

4. The top floor number (F6-00) and bottom floor number (F6-01) are set correctly.

5. The system is not in the fault alarm state. If there is a fault at the moment, press 

to reset the fault.

6. Then set F7-26 to 1 on the operation panel or set F7 to 1 on the keypad of the SSL-7000-A1, and start shaft auto-tuning.

Note

For shaft auto-tuning when there are only two floors, the elevator needs to run to below the bottom leveling position, that is, the leveling sensor is disconnected from the leveling plate. There is no such requirement when there are multiple floors.

2.4.3 Riding Comfort

The riding comfort is an important factor of the elevator's overall performance. Improper installation of mechanical parts and improper parameter settings will cause discomfort. Enhancing the riding comfort mainly involves adjustment of the controller output and the elevator's mechanical construction.

- **Controller Output**

The parameters that may influence the riding comfort are described in this part.

Function Code	Parameter Name	Setting Range	Default	Description
F1-10	Magnetizing current	0.01–300.00	**0.00 A**	Increasing the value can improve the loading capacity of the asynchronous motor.
F2-00	Speed loop proportional gain KP1	0–100	40	F2-00 and F2-01 are the PI regulation parameters when the running frequency is lower than F2-02 (Switchover frequency 1). F2-03 and F2-04 are the PI regulation parameters when the running frequency is higher than F2-02 (Switchover frequency 2). The regulation parameters between F2-02 and F2-04 are the weighted average value of F2-00 & F2-01 and F2-03 & F2-04.
F2-01	Speed loop integral time TI1	0.01–10.00s	0.60s	
F2-02	Switchover frequency 1	0.00 to F2-05	2.00 Hz	
F2-03	Speed loop proportional gain KP2	0–100	35	
F2-04	Speed loop integral time TI2	0.01–10.00s	0.80s	
F2-05	Switchover frequency 2	F2-02 to F0-06	5.00 Hz	

For a faster system response, increase the proportional gain and reduce the integral time. Be aware that a fast system response causes system oscillation.

The recommended regulating method is as follows:

If the default setting cannot satisfy the requirements, make slight regulation. Decrease the proportional gain first to ensure that the system does not oscillate. Then decrease the integral time to ensure fast responsiveness and small overshoot.

If both F2-02 (Switchover frequency 1) and F2-05 (Switchover frequency 2) are set to 0, only F2-03 and F2-04 are valid.

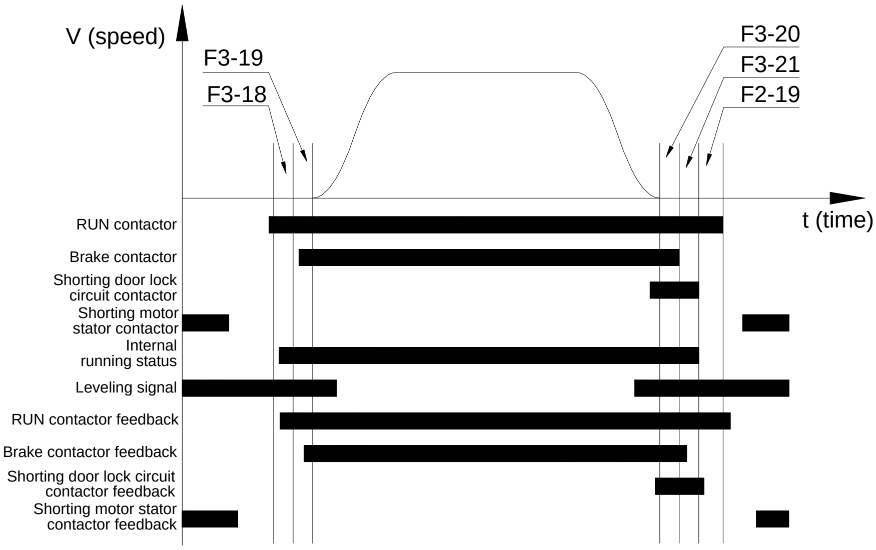
Function Code	Parameter Name	Setting Range	Default	Description
F2-06	Current loop proportional gain	10–500	60	F2-06 and F2-07 are the current loop adjustment parameters in the vector control algorithm.
F2-07	Current loop integral gain	10–500	30	

The optimum values of these two parameters are obtained during motor auto-tuning, and you need not modify them. Appropriate setting of the parameters can restrain jitter during running and have obvious effect on the riding comfort.

Function Code	Parameter Name	Setting Range	Default	Description
F2-20	Current filter coefficient	0.00–40.00s	**0.00**	It can reduce the lower-frequency vertical jitter during running.
F2-22	Startup acceleration time	0.000–1.500s	0.000s	It can reduce the terrace feeling at startup caused by the breakout friction of the guide rail.
F3-00	Startup speed	0.000–0.030 m/s	0.000 m/s	
F3-01	Startup holding time	0.000–0.500s	0.000s	
F3-18	Zero-speed control time at startup	0.000–1.000s	0.200s	It specifies the time from output of the RUN contactor to output of the brake contactor, during which the controller performs excitation on the motor and outputs zero-speed current with large startup torque.
F3-19	**Brake release delay**	0.000–1.500s	0.200s 0.600s	It specifies the time from the moment when the system sends the brake release command to the moment when the brake is completely released, during which the system retains the zero-speed torque current output.
F3-20	**Zero-speed control time at end**	0.000–1.000s	0.300s	It specifies the zero-speed output time when the running curve ends.
F3-21	**Brake apply delay**	0.200–1.500s	0.200s	It specifies the time from the moment when the system sends the brake apply command to the moment when the brake is completely applied, during which the system retains the zero-speed torque current

				output.

Figure 2-6 Running time sequence



The release time of the brakes varies according to the types and the response time of the brakes is greatly influenced by the ambient temperature. A very high brake coil temperature slows the brake responsiveness. Thus, when the riding comfort at startup or stop cannot be improved by adjusting zero servo or load cell compensation parameters, appropriately increase the values of F3-19 and F3-21 to check whether the brake release time influences the riding comfort.

Function Code	Parameter Name	Setting Range	Default	Remarks
F2-11	No-load-cell startup	0: Disabled 1: Enabled	0	These are zero-servo regulating parameters (Automatic pre-torque compensation).
F2-12	Zero servo speed KP	0.00–2.00	0.50	
F2-13	Zero servo speed KI	0.00–2.00	0.60	
F2-14	Zero servo current Kp1	10–1000	60	
F2-15	Zero servo current Ki1	10–1000	30	

The system automatically adjusts the compensated torque at startup. It is applicable to all types of encoder (ERN1387 as the best).

a) Set F2-11 to 1 to enable no-load-cell startup.

b) Gradually increase the value of F2-12 (Zero servo speed KP) to ensure that the motor does not oscillate.

c) Gradually increase the value of F2-13 (Zero servo speed KI) if increasing the value of F2-12 still cannot meet the requirements.

d) If the motor noise is large at no-load-cell startup, decrease the value of zero servo current parameter (F2-14/F2-15).

Function Code	Parameter Name	Setting Range	Default	Remarks
F8-01	Pre-torque selection	0: Pre-torque invalid 1: Load-cell pre-torque compensation	0	These are pre-torque regulating parameters.
F8-02	Pre-torque offset	0.0%–100.0%	50.0%	
F8-03	Drive gain	0.00–2.00	0.60	
F8-04	Brake gain	0.00–2.00	0.60	

When F8-01 is set to 1 (Load cell pre-torque compensation), the system with a load cell pre-outputs the torque matched the load to ensure the riding comfort of the elevator.

- Motor driving state: full-load up, no-load down
- Motor braking state: full-load down, no-load up

F8-02 (Pre-torque offset) is actually the elevator balance coefficient, namely, the percentage of the car load to the rated load when the car and counterweight are balanced.

F8-03 (Drive gain) or F8-04 (Brake gain) scales the elevator's present pre-torque coefficient when the motor runs at the drive or brake side. If the gain set is higher, then the calculated value of startup pro-torque compensation is higher. The controller identifies the braking or driving state according to the load cell signal and automatically calculates the required torque compensation value.

When an analog device is used to measure the load, these parameters are used to adjust the elevator startup. The method of adjusting the startup is as follows:

- In the driving state, increasing the value of F8-03 could reduce the rollback during the elevator startup, but a very high value could cause car lurch at start.
- In the braking state, increasing the value of F8-04 could reduce the jerk in command direction during the elevator startup, but a very high value could cause car lurch at start.

■ **Mechanical Construction**

The mechanical construction affecting the riding comfort involves installation of the guide rail, guide shoe, steel rope, and brake, balance of the car, and the resonance caused by the car, guild rail and motor. For asynchronous motor, abrasion or improper installation of the gearbox may arouse poor riding comfort.

1. Installation of the guide rail mainly involves the verticality and surface flatness of the guide rail, smoothness of the guide rail connection and parallelism between two guide rails (including guide rails on the counterweight side).
2. Tightness of the guide shoes (including the one on the counterweight side) also influences the riding comfort. The guide shoes must not be too loose or tight.
3. The drive from the motor to the car totally depends on the steel rope. Large flexibility of the steel rope with irregular resistance during the car running may cause curly oscillation of the car. In addition, unbalanced stress of multiple steel ropes may cause the car to jitter during running.
4. The riding comfort during running may be influenced if the brake arm is installed too tightly or released incompletely.
5. If the car weight is unbalanced, it will cause uneven stress of the guide shoes that connect the car and the guide rail. As a result, the guide shoes will rub with the guide rail during running, affecting the riding comfort.
6. For asynchronous motor, abrasion or improper installation of the gearbox may also affect the riding comfort.
7. Resonance is an inherent character of a physical system, related to the material and quality of system components. If you are sure that the oscillation is caused by resonance, reduce the resonance by increasing or decreasing the car weight or counterweight and adding resonance absorbers at connections of the components (for example, place rubber blanket under the motor).

2.4.4 Door Machine Controller Commissioning

Correlation of the door machine controller and the elevator controller is that the SSL7020 outputs door open/close command and the door machine controller feeds back the door open/close limit signal.

After commissioning and installation of the door machine controller are complete, check whether the wiring is correct and whether the door open/close limit signals are consistent with the default setting. To perform the door machine controller commissioning, do as follows:

1. In the terminal control mode of the door machine controller, manually short the door open relay output terminal BM/B1 and the door close relay output terminal BM/B2 on the SSL7020, and observe whether the door machine can open and close correspondingly. If the door machine cannot

act properly, check whether BM/B1 and BM/B2 are wrongly connected to the input terminals of the door machine controller and whether commissioning of the door machine controller is complete.

2. After ensuring that control of door open/close is normal, check whether the door open/close signal feedback from the door machine controller is normal.

   a. Check the NO/NC states of the door input signals by observing the input indicators on the SSL7020, as listed in the following table.

Table 2-4 NO/NC state of the door input signals

Door State	Signal Input Point	NO Input Signal		NC Input Signal	
		Indicator State	**FL-00 Setting**	**Indicator State**	**FL-00 Setting**
Door open limit	X3 (door open limit 1)	When the signal is active, the corresponding input indicator is ON.	Bit2 = 1	When the signal is active, the corresponding input indicator is OFF.	Bit2 = 0
	X4 (door open limit 2)		Bit3 = 1		Bit3 = 0
Door close limit	X5 (door close limit 1)		Bit4 = 1		Bit4 = 0
	X6 (door close limit 2)		Bit5 = 1		Bit5 = 0

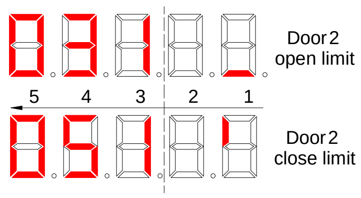
For details on the setting of FL-00, see the description of FL-00 in Chapter 3.

b. Check whether the door open/close limit signal received by the system is correct.

As shown in the following figure which is part of display of parameter FU-26 on the operation panel.

If the display is the same as the following figure, the door open and close limit signals from the controller are correct.

Figure 2-7 Fu-26 Door open/close limit monitoring

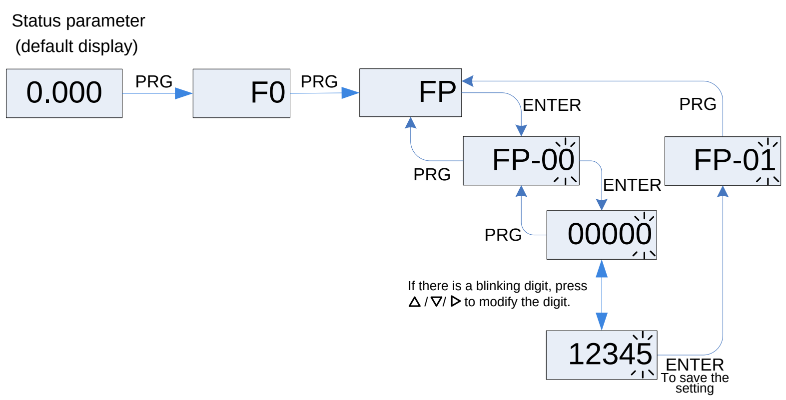


2.4.5 Password Setting

The SLEC7000 provides the parameter password protection function.

Here gives an example of changing the password into 12345 (indicates the blinking digit), as shown in the following figure.

Figure 2-8 Example of changing the password



- After you set the user password (set FP-00 to a non-zero value), the system requires user password authentication (the system displays "------") when you press PRG. In this case, you can modify the function code parameters only after entering the password correctly.

- For factory parameters, you also need to enter the factory password.
- Do not try to modify the factory parameters. If these parameters are set improperly, the system may be unstable or abnormal.
- In the password protection unlocked state, you can change the password at any time. The last input number will be the user password.
- If you want to disable the password protection function, enter the correct password and then set FP-00 to 0. If FP-00 is a non-zero value at power-on, the parameters are protected by the password.

Remember the password you set. Otherwise, the system cannot be unlocked.

Chapter 3 Function Code Table

The modification property of the parameters includes three types, described as follows:

"☆ ": The parameter can be modified when the controller is in either stop or running state.

"★ ": The parameter cannot be modified when the controller is in the running state.

"●": The parameter is the actually measured value and cannot be modified.

Function Code	Parameter Name	Setting Range	Default	Unit	Property
Group F0: Basic parameters					
F0-00	Control mode	0: Sensorless vector control (SVC) 1: Closed-loop vector control (CLVC) 2: Voltage/Frequency (V/F) control	1	-	★
F0-01	Command source selection	0: Operation panel control 1: Distance control	1	-	★
F0-02	Running speed under operation panel control	0.050 to F0-04	0.050	m/s	☆
F0-03	Maximum running speed	0.100 to F0-04	1.600	m/s	★
F0-04	Rated elevator speed	0.100 to 4.000	1.600	m/s	★
F0-05	Direction selection	0: Direction unchanged 1: Direction reversed	0	-	★
F0-07	Carrier frequency	2.0–16.0	8.0	kHz	☆
Group F1: Motor parameters					
F1-00	Security password	0–65535	01000	-	●
F1-01	Rated motor power	1.1–132.0	Model dependent	kW	★
F1-02	Rated motor voltage	50–600	380	V	★
F1-03	Rated motor current	0.00–655.00	25.00	A	★
F1-04	Rated motor frequency	0.00–99.00	50.00	Hz	★
F1-05	Rated motor rotational speed	0–3000	1460	RPM	☆
F1-06	Stator resistance	0.000–65.000	0.000	Ω	☆
F1-07	Rotor resistance	0.000–65.000	0.000	Ω	☆
F1-08	Leakage inductance	0.00–650.00	2.34	mH	☆
F1-09	Mutual inductance	0.0–3000.0	66.6	mH	☆
F1-10	Magnetizing current	0.01–650.00	10.70	A	☆
F1-11	Auto-tuning mode	0: No operation 1: With-load auto-tuning 2: No-load auto-tuning 3: Parameter auto-tuning	0	-	★
F1-12	Auto-tuning setting	Bit1: Angle-free auto-tuning	0	-	★
F1-13	Auto-tuning current	30–150	60	%	★

Function Code	Parameter Name	Setting Range	Default	Unit	Property
F1-15	Shaft Q inductance (torque)	0.00–650.00	3.00	mH	★
F1-16	Shaft D inductance (excitation)	0.00–650.00	3.00	mH	★
F1-17	Back EMF	0–65535	0	-	★
F1-25	Motor type	0: Asynchronous motor 1: Synchronous motor	1	-	★
Group F2: **Vector control parameters**					
F2-00	Speed loop proportional gain Kp1	1–100	40	-	☆
F2-01	Speed loop integral time Ti1	0.01–10.00	0.60	s	☆
F2-02	Switchover frequency 1	0.00 to F2-05	2.00	Hz	☆
F2-03	Speed loop proportional gain Kp2	1–100	35	-	☆
F2-04	Speed loop integral time Ti2	0.01–10.00	0.80	s	☆
F2-05	Switchover frequency 2	F2-02 to F1-04	5.00	Hz	☆
F2-06	Current loop Kp1 (torque)	10–1000	60	-	☆
F2-07	Current loop Ki1 (torque)	10–1000	30	-	☆
F2-08	Torque upper limit	0.0–200.0	150.0	%	☆
F2-09	Current loop Kp2 (excitation)	10–1000	60	-	☆
F2-10	Current loop Ki2 (excitation)	10–1000	30	-	☆
F2-11	No-load-cell startup	0: Disabled 1: Enabled	0	-	★
F2-12	Zero servo speed loop Kp	1–100	35	-	☆
F2-13	Zero servo speed loop Ki	0.01–10.00	0.80	-	☆
F2-14	Zero servo current loop Kp1 (torque)	10–1000	60	-	☆
F2-15	Zero servo current loop Ki1 (torque)	10–1000	30	-	☆
F2-16	Zero servo current loop Kp2 (excitation)	10–1000	60	-	☆
F2-17	Zero servo current loop Ki2(excitation)	10–1000	30	-	☆
F2-18	Torque acceleration time	0–500	1	ms	★
F2-19	Torque deceleration time	0–500	350	ms	★
F2-20	Current filter coefficient	0.00–40.00	0.00	-	★
F2-21	Zero servo rollback	0–9999	0	Pulses	●

Function Code	Parameter Name	Setting Range	Default	Unit	Property
	index				
F2-22	Startup acceleration time	0.001–1.500	0	s	★
Group F3: **Running control parameters**					
F3-00	Startup speed	0.000–0.030	0.000	m/s	★
F3-01	Startup holding time	0.000–1.500	0.150	s	★
F3-02	**Acceleration rate**	0.300–1.300	0.600	m/s^2	★
F3-03	**Acceleration start jerk time**	0.800–3.000	2.500	s	★
F3-04	**Acceleration end jerk time**	0.800–3.000	2.500	s	★
F3-05	Deceleration rate	0.300–1.300	0.600	m/s^2	★
F3-06	Deceleration end jerk time	0.800–3.000	2.500	s	★
F3-07	Deceleration start jerk time	0.800–3.000	2.500	s	★
F3-08	**Special deceleration rate**	0.800–1.500	0.900	m/s^2	★
F3-09	Pre-deceleration distance	0–50.0	0	mm	★
F3-10	Re-leveling speed	0.040–0.080	0.040	m/s	★
F3-11	Inspection speed	0.080–0.630	0.250	m/s	★
F3-12	Low-speed self-rescue speed	0.080 to F3-11	0.100	m/s	★
F3-13	Terminal floor verification speed	0.100 to F0-04	0.500	m/s	★
F3-14	Emergency evacuation function selection	0–65535	32	-	★
F3-15	Emergency evacuation speed	0.080–0.500	0.080	m/s	★
F3-16	Emergency evacuation acceleration rate	0.500–2.000	0.500	m/s^2	★
F3-17	High-speed emergency evacuation times	0–10	0	-	★
F3-18	**Zero-speed control time at startup**	0.000–1.000	0.200	s	★
F3-19	**Brake release delay**	0.000–1.500	0.600	s	★
F3-20	**Zero-speed control time at end**	0.000–1.000	0.300	s	★
F3-21	**Brake apply delay**	0.200–1.500	0.200	s	★
Group F4: Floor parameters					
F4-00	Leveling adjustment	0–60	30	mm	★
F4-01	Current floor	F6-01 to F6-00	1	-	★

Function Code	Parameter Name	Setting Range	Default	Unit	Property
F4-02	High byte of current floor position	0–65535	0	Pulses	★
F4-03	Low byte of current floor position	0–65535	0	Pulses	★
F4-04	Length 1 of leveling plate	0–65535	0	mm	★
F4-05	Length 2 of leveling plate	0–65535	0	mm	★
F4-06	Leveling delay	0–80	28	ms	★
F4-07	Down leveling adjustment	0–60	30	mm	★
F4-10	High byte of floor height 1	0–65535	0	Pulses	★
F4-11	Low byte of floor height 1	0–65535	0	Pulses	★
F4-12	High byte of floor height 2	0–65535	0	Pulses	★
F4-13	Low byte of floor height 2	0–65535	0	Pulses	★
F4-14	High byte of floor height 3	0–65535	0	Pulses	★
F4-15	Low byte of floor height 3	0–65535	0	Pulses	★
F4-16	High byte of floor height 4	0–65535	0	Pulses	★
F4-17	Low byte of floor height 4	0–65535	0	Pulses	★
F4-18	High byte of floor height 5	0–65535	0	Pulses	★
F4-19	Low byte of floor height 5	0–65535	0	Pulses	★
Floor height 6 to floor height 53					
F4-116	High byte of floor height 54	0–65535	0	Pulses	★
F4-117	Low byte of floor height 54	0–65535	0	Pulses	★
F4-118	High byte of floor height 55	0–65535	0	Pulses	★
F4-119	Low byte of floor height 55	0–65535	0	Pulses	★
Group F5: Terminal function parameters					
F5-00	Load cell input selection	0: Invalid 1: SSL7020 digital input 2: SSL7020 analog input 3: SSL-7000-A1 analog input	2	-	★

Function Code	Parameter Name	Setting Range	Default	Unit	Property
F5-01	X1 function selection	000： Invalid 016：Up slow-down 1 signal	116		★
F5-02	X2 function selection	000： Invalid 018：Up slow-down 2 signal	118		★
F5-03	X3 function selection	Multi-function input	000		★
F5-04	X4 function selection	000： Invalid 017：Down slow-down 1 signal	117		★
F5-05	X5 function selection	000：Invalid 019：Down slow-down 2 signal	119		★
F5-06	X6 function selection	Multi-function input	000		★
F5-07	X7 function selection	000：Invalid 001： Up leveling signal	001		★
F5-08	X8 function selection	000：Invalid 002：Down leveling signal	002		★
F5-09	X9 function selection	000：Invalid 003：Door zone signal	003		★
F5-10	X10 function selection	000：Invalid 008：Inspection signal	108		★
F5-11	X11 function selection	000：Invalid 009：Inspection up signal	009		★
F5-12	X12 function selection	000：Invalid 010：Inspection down signal	010		★
F5-13	X13 function selection	000：Invalid 006：Brake contactor feedback signal	106		★
F5-14	X14 function selection	000：Invalid 030： Shorting PMSM stator feedback signal	030		★
F5-15	X15 function selection	000：Invalid 007：Brake feedback signal	107		★
F5-16	X16 function selection	Multi-function input	000		★
F5-17	X17 function selection	Multi-function input	000		★
F5-18	X18 function selection	000：Invalid 027：Emergency evacuation signal	127		★
F5-19	X19 function selection	000：Invalid 022：Shorting door lock circuit contactor feedback	022		★
F5-20	X20 function selection	Multi-function input	000		★
F5-21	X21 function selection	000：Invalid 031：Left brake detection switch feedback signal	131		★
F5-22	X22 function selection	000：Invalid 031：Right brake detection switch	131		★

Function Code	Parameter Name	Setting Range	Default	Unit	Property
		feedback signal			
F5-23	X23 function selection	000：Invalid 032： Motor overheat signal	132		★
F5-24	X24 function selection	Multi-function input	014		★
F5-25	X25 function selection	000：Invalid 012：Up limit signal	112		★
F5-26	X26 function selection	000：Invalid 013：Down limit signal	113		★
F5-27	X27 function selection	000：Invalid 011：Fire emergency signal	011		★
F5-28	X28 function selection	Multi-function input	034		★

Multi-function output：

00	Invalid	01	Up leveling signal	02	Down leveling signal	03	Door zone signal
04	Safety circuit feedback signal	05	Door lock circuit feedback signal	06	Brake contactor feedback signal	07	Brake feedback signal
08	Inspection signal	09	Inspection up signal	10	Inspection down signa	11	Fire emergency signal
12	Up limit signal	13	Down limit signal	14	Overload signal	15	Full-load signal
16	Up slow-down 1 signal	17	Down slow-down 1 signal	18	Up slow-down 2 signal	19	Down slow-down 2 signal
20	Up slow-down 3 signal	21	Down slow-down 3 signal	22	Shorting door lock circuit contactor feedback	23	Firefighter switch signal
24	Door machine 1 light curtain signal	25	Door machine 2 light curtain signal	26	Door machine 1 safety edge signal	27	Emergency evacuation signal
28	Elevator lock signal	29	Door machine 2 safety edge signal	30	Shorting PMSM stator feedback signal	31	Brake detection switch feedback signal
32	Motor overheat signal	33	VIP signal	35	Security signal	42	Back door selection signal

Note: The value is a three-digit number. The hundred's digit indicates the NO/NC state of

Function Code	Parameter Name	Setting Range	Default	Unit	Property
the signal: 0 indiates that the signal is NO; 1 indicates that the signal is NC.					
F5-29	X29 function selection	0：Invalid 4：Safety circuit signal	4		★
F5-30	X30 function selection	0：Invalid 5：Door lock circuit signal	5		★
F5-31	X31 function selection	0：Invalid 5：Door lock circuit signal	5		★
F5-32	Y1 function selection	0：Invalid 1：Main contactor control	1		★
F5-33	Y2 function selection	0：Invalid 2：Brake contactor control	2		★
F5-34	Y3 function selection	Multi-function output	16		★
F5-35	Y4 function selection	0：Invalid 12：Shorting door lock circuit contactor control	12		★
F5-36	Y5 function selection	Multi-function output	0		★
F5-37	Y6 function selection	0：Invalid 13：Emergency evacuation automatic switchover	13		★
F5-38	Y7 function selection	Multi-function output	23		★
F5-39	Y8 function selection	Multi-function output	3		★
F5-40	Y9 function selection	Multi-function output	4		★

Multi-function output：

0	Invalid	1	Main contactor control	2	Brake contactor control
3	Shorting door lock circuit contactor control	4	Fire emergency floor arrival signal feedback	5	Door machine 1 open
6	Door machine 1 close	7	Door machine 2 open	8	Door machine 2 close
10	Fault state	11	Running monitor	12	Shorting PMSM stator contactor control
13	Emergency evacuation automatic switchover	15	Emergency buzzer control	16	Higher-voltage startup of brake
17	Elevator running in up direction	18	Lamp/Fan running	19	Medical sterilization
20	Non-door zone stop	23	Electric lock output	22	Non-service state output

Function Code	Parameter Name	Setting Range	Default	Unit	Property
Group F6: Elevator logic parameters					
F6-00	Top floor	F6-01 to 56	9	-	★
F6-01	Bottom floor	1 to F6-00	1	-	★
F6-02	Parking floor	F6-01 to F6-00	1	-	★
F6-03	Fire emergency floor	F6-01 to F6-00	1	-	★
F6-04	Elevator lock floor	F6-01 to F6-00	1	-	★
F6-05	Service floors 1	0–65535	65535	-	★
F6-06	Service floors 2	0–65535	65535	-	★
F6-07	Service floors 3	0–65535	65535	-	★

Function Code	Parameter Name	Setting Range	Default	Unit	Property
F6-08	Service floors 4	0–65535	65535	-	★
F6-09	Elevator function selection 1	Bit0: Disability function Bit2: Re-leveling function Bit3: Door pre-open function Bit6: Down collective selective function Bit7: Fault auto-reset function Bit8: Time-based service floor function Bit11: Car call deletion Bit13: Timed elevator lock Bit14: Arrival gong disabled at night Bit15: Reserved	34816	-	★
F6-10	Elevator function selection 2	Bit0: Door open/close holding at door open/close limit Bit2: Dooropen/close command cancelled immediately at door open/close limit Bit5: Forced door close Bit12: Car call assisted command in single door used as disability function Bit14: Car call command folding	18	-	★
F6-22	Elevator lock start time	00.00–23.59	0	-	★
F6-23	Elevator lock end time	00.00–23.59	0	-	★
F6-24	Attendant function selection	Bit0: Calls cancelled after entering attendant state Bit1: Not responding to hall calls Bit2: Attendant auto exit Bit3: Jog door close Bit4: Automatic door close Bit5: Buzzer tweeting	0	-	★
F6-25	Attendant/Automatic switchover time	0–200	0	s	★
F6-26	Emergency evacuation lasting time	30–600	45	s	★
F6-27	Emergency evacuation function selection	Bit0: Door 1 valid Bit1: Door 2 valid Bit2: Emergency running time protection Bit3: Manual funciton Bit4: Emergency buzzer tweet Bit5: SSL7040 display Bit6: Switching over shorting stator braking mode to controller drive Bit7: Mode of shorting stator braking mode switched over to controller drive Bit8: Emergency evacuation exit mode	0	-	★
F6-28	Inspection function seslection	Bit0: Inspection fire emergency prompt Bit1: Door lock disconnected at	0	-	★

Function Code	Parameter Name	Setting Range	Default	Unit	Property
		inspection switched over to normal running Bit5: Power-on and door open once at inspection switched over to normal running			
F6-30	VIP function selection	Bit0: VIP1 function Bit1: VIP2 function Bit2: VIP enabled by hall call Bit3: VIP enabled by button Bit4: VIP enabled by terminal at any floor Bit5: VIP enabled by button at any floor Bit6: VIP enabled by car call Bit7: Security floor automatically enabled Bit8: Number of VIP car calls limited Bit9: VIP auto exit Bit13: Button holding Bit14: Automatic door close	0	-	★
F6-31	VIP floor 1	0 to F6-00	0	-	★
F6-32	VIP floor2	0 to F6-00	0	-	★
F6-33	VIP auto exit time	0–200	0	s	★
F6-35	Start time of time-based floor service 1	00.00–23.59	0	-	☆
F6-36	End time of time-based floor service 1	00.00–23.59	0	-	☆
F6-37	Service floor 1 of time-based floor service 1	0–65535	65535	-	☆
F6-38	Service floor 2 of time-based floor service 1	0–65535	65535	-	☆
F6-39	Service floor 3 of time-based floor service 1	0–65535	65535	-	☆
F6-40	Service floor 4 of time-based floor service 1	0–65535	65535	-	☆
F6-41	Start time of time-based floor service 2	00.00–23.59	0	-	☆
F6-42	End time of time-based floor service 2	00.00–23.59	0	-	☆
F6-43	Service floor 1 of time-based floor service 2	0–65535	65535	-	☆
F6-44	Service floor 2 of time-based floor service 2	0–65535	65535	-	☆

Function Code	Parameter Name	Setting Range	Default	Unit	Property
F6-45	Service floor 3 of time-based floor service 2	0–65535	65535	-	☆
F6-46	Service floor 4 of time-based floor service 2	0–65535	65535	-	☆
F6-47	Start time of time-based floor service 3	00.00–23.59	0	-	☆
F6-48	End time of time-based floor service 3	00.00–23.59	0	-	☆
F6-49	Service floor 1 of time-based floor service 3	0–65535	65535	-	☆
F6-50	Service floor 2 of time-based floor service 3	0–65535	65535	-	☆
F6-51	Service floor 3 of time-based floor service 3	0–65535	65535	-	☆
F6-52	Service floor 4 of time-based floor service 3	0–65535	65535	-	☆
Group F7: Test function parameters					
F7-00	Car call floor registered	0 to F6-00	0	-	☆
F7-01	Up call floor registered	0 to F6-00	0	-	☆
F7-02	Down call floor registered	0 to F6-00	0	-	☆
F7-03	Random running times	0–60000	0	-	☆
F7-04	Hall call enabled	0: Yes 1: No	0	-	☆
F7-05	Door open enabled	0: Yes 1: No	0	-	☆
F7-06	Overload function	0: Disabled 1: Enabled	0	-	☆
F7-07	Limit switch	0: Enabled 1: Disabled	0	-	☆
F7-08	Time interval of random running	0–1000	0	s	☆
F7-09	Accumulative consumed electricity	0–65535	0	kW	●
F7-10	Accumulative feedback electricity	0–65535	0	kW	●
F7-26	Commissioning function selection	0: No operation 1: Shaft auto-tuning	0	-	★
Group F8: Auxiliary logic parameters					

Function Code	Parameter Name	Setting Range	Default	Unit	Property
F8-00	Load for load cell auto-tuning	0–100	0	%	★
F8-01	Pre-torque selection	0: Invalid 1: Valid	0	-	★
F8-02	Pre-torque offset	0.0–100.0	50.0	%	★
F8-03	Drive gain	0.00–2.00	0.60	-	★
F8-04	Brake gain	0.00–2.00	0.60	-	★
F8-05	Current load	0–255	0	-	●
F8-06	Car no-load load	0–255	0	-	★
F8-07	Car full-load load	0–255	100	-	★
F8-08	Anti-nuisance function	Bit0: Nuisance judged by load cell Bit1: Nuisance judged by door close Bit2: Nuisance judged by light-load signal	0	-	★
F8-09	Rated elevator log	300–9999	1000	kg	★
F8-10	Program control	Bit13: Hide non-commonly used parameters	0	-	★
F8-14	Local address	0–127	1	-	★
F8-18	Overload threshold	100–130	110	%	★
F8-19	Full load threshold	70–110	80	%	★
F8-20	Light load threshold	10–50	30	%	★
F8-21	Arrival gong advance time	0–10.0	1.0	s	★
F8-22	Door open limit delay	0–2000	1000	ms	★
F8-24	Command 2 start address	0–56	0	-	★
Group F9: Time parameters					
F9-00	Idle time before returning to base floor	0–240	10	min	★
F9-01	Time for fan and lamp to be turned off	0–240	2	min	★
F9-02	Motor running time limit	0–45	45	s	★
F9-03	Accumulative power-on time (hour)	0–65535	0	h	●
F9-04	Accumulative running time	0–65535	0	h	●
F9-05	Accumulative power-on time (day)	0–65535	0	day	●
F9-06	High byte of running times	0–9999	0	-	●
F9-07	Low byte or running times	0–9999	0	-	●
F9-12	Clock: year	2010–2100	2010	YYYY	☆

Function Code	Parameter Name	Setting Range	Default	Unit	Property
F9-13	Clock: month and day	1.01–12.31	301	MM.DD	☆
F9-14	Clock: hour and minute	0–23.59	0	HH.MM	☆
Group FA: Auxiliary parameter					
FA-00	Security password for group FA	0–65535	01000	-	●
FA-01	Display in running state	1–65535	65535	-	☆
FA-02	Display in stop state	1–65535	4095	-	☆
FA-03	Product SN	1–7000	7000	-	●
FA-04	Software version 1 (SSL7020)	0–65535	0	-	●
FA-05	Software version 2 (SSL-7000-A1)	0–65535	0	-	●
FA-06	Software version 3 (Drive board)	0–65535	0	-	●
FA-07	Heatsink temperature	0–100	0	°C	●
FA-09	Protection function selection	Bit0: Overload protection Bit1: Protection at output phase loss Bit2: Over-modulation function Bit3: Protection at input phase loss Bit4: Pulse-to-pulse current limit	3	-	★
FA-10	Overload protection coefficient	0.50–10.00	1.00	-	★
FA-11	Overload pre-warning coefficient	50–100	80	%	★
FA-15	Program control	Bit0: Super short floor function Bit1: Up slow-down not reset for super short floor Bit2: Down slow-down not reset for super short floor Bit3: Cancelling monitor of slow-down at leveling Bit4: Detecting slow-down stuck Bit5: Cancelling shaft auto-tuning 45s detetction Bit6: Leveling adjustment function	0	-	★
FA-37			20		
Group FB: Door function parameters					
FB-00	Number of door machine(s)	1–2	1	-	★
FB-01	Opposite door control mode	0–65535	0	-	★
FB-02	Service floors 1 of door machine 1	0–65535	65535	-	☆
FB-03	Service floors 2 of door machine 1	0–65535	65535	-	☆
FB-04	Service floors 3 of door	0–65535	65535	-	☆

Function Code	Parameter Name	Setting Range	Default	Unit	Property
	machine 1				
FB-05	Service floors 4 of door machine 1	0–65535	65535	-	☆
FB-06	Service floors 1 of door machine 2	0–65535	65535	-	☆
FB-07	Service floors 2 of door machine 2	0–65535	65535	-	☆
FB-08	Service floors 3 of door machine 2	0–65535	65535	-	☆
FB-09	Service floors 4 of door machine 2	0–65535	65535	-	☆
FB-10	Door open protection time	5–99	10	s	★
FB-11	Forced door close time	5–99	15	s	★
FB-12	Door close protection time	5–99	15	s	★
FB-13	Door re-open times	0–20	0	-	★
FB-14	Door state of standby elevator	0: Closing the door as normal at base floor 1: Waiting with door open at base floor 2: Waiting with door open at each floor	0	-	★
FB-15	Door open holding time for hall call	1–30	5	s	★
FB-16	Door open holding time for car call	1–30	3	s	★
FB-17	Door open holding time at base floor	1–30	10	s	★
FB-18	Door open delay	10–30000	30	s	★
FB-19	Special door open holding time	10–1000	30	s	★
FB-20	Manual door open holding time	1–60	10	s	★
Group FC: Fault parameters					
FC-00	Designated fault code	0–99	0	-	●
FC-01	20th fault code	0–6299	0	-	●
FC-02	20th fault subcode	0–65535	0	-	●
FC-03	20th fault month and day	0–1231	0	MM.DD	●
FC-04	20th fault time	0–2359	0	HH.MM	●
FC-05	20th fault information	0–65535	0	-	●
FC-06	19th fault code	0–6299	0	-	●
FC-07	19th fault subcode	0–65535	0	-	●
FC-08	19th fault month and day	0–1231	0	MM.DD	●

Function Code	Parameter Name	Setting Range	Default	Unit	Property
FC-09	19th fault time	0–2359	0	HH.MM	●
FC-10	19th fault information	0–65535	0	-	●
FC-11	18th fault code	0–6299	0	-	●
FC-12	18th fault subcode	0–65535	0	-	●
FC-13	18th fault month and day	0–1231	0	MM.DD	●
FC-14	18th fault time	0–2359	0	HH.MM	●
FC-15	18th fault information	0–65535	0	-	●
FC-16	17th fault code	0–6299	0	-	●
FC-17	17th fault subcode	0–65535	0	-	●
FC-18	17th fault month and day	0–1231	0	MM.DD	●
FC-19	17th fault time	0–2359	0	HH.MM	●
FC-20	17th fault information	0–65535	0	-	●
FC-21	16th fault code	0–6299	0	-	●
FC-22	16th fault subcode	0–65535	0	-	●
FC-23	16th fault month and day	0–1231	0	MM.DD	●
FC-24	16th fault time	0–2359	0	HH.MM	●
FC-25	16th fault information	0–65535	0	-	●
FC-26	15th fault code	0–6299	0	-	●
FC-27	15th fault subcode	0–65535	0	-	●
FC-28	15th fault month and day	0–1231	0	MM.DD	●
FC-29	15th fault time	0–2359	0	HH.MM	●
FC-30	15th fault information	0–65535	0	-	●
FC-31	14th fault code	0–6299	0	-	●
FC-32	14th fault subcode	0–65535	0	-	●
FC-33	14th fault month and day	0–1231	0	MM.DD	●
FC-34	14th fault time	0–2359	0	HH.MM	●
FC-35	14th fault information	0–65535	0	-	●
FC-36	13th fault code	0–6299	0	-	●
FC-37	13th fault subcode	0–65535	0	-	●
FC-38	13th fault month and day	0–1231	0	MM.DD	●
FC-39	13th fault time	0–2359	0	HH.MM	●
FC-40	13th fault information	0–65535	0	-	●
FC-41	12th fault code	0–6299	0	-	●
FC-42	12th fault subcode	0–65535	0	-	●
FC-43	12th fault month and day	0–1231	0	MM.DD	●
FC-44	12th fault time	0–2359	0	HH.MM	●
FC-45	12th fault information	0–65535	0	-	●
FC-46	11th fault code	0–6299	0	-	●

Function Code	Parameter Name	Setting Range	Default	Unit	Property
FC-47	11th fault subcode	0–65535	0	-	●
FC-48	11th fault month and day	0–1231	0	MM.DD	●
FC-49	11th fault time	0–2359	0	HH.MM	●
FC-50	11th fault information	0–65535	0	-	●
Group Fd: Parallel/Group control parameters					
Fd-00	Number of elevators in parallel/group mode	1–8	1	-	★
Fd-01	Elevator No.	1–8	1	-	★
Fd-02	Group control mode	0: Parallel control 1: Group control of destination dispatch 2: Group control of traditional mode	-	-	-
Fd-03	Group control function selection	Bit0: Dispersed waiting Bit1: Dispersed waiting mode selection Bit3: Main/Auxiliary elevators automatic switchover Bit4: Cancelling long-time waiting	-	-	-
Group FE: Display setting					
FE-01	Floor 1 display	00: Display "0" 01: Display "1" 02: Display "2" 03: Display "3" 04: Display "4" 05: Display "5" 06: Display "6" 07: Display "7" 08: Display "8" 09: Display "9" 10: Display "A" 11: Display "B" 12: Display "G" 13: Display "H" 14: Display "L" 15: Display "M" 16: Display "P" 17: Display "R" 18: Display "-" 19: No display 20: Display "12" 21: Display "13" 22: Display "23" 23: Display "C" 24: Display "D" 25: Display "E" 26: Display "F" 27: Display "I" 28: Display "J" 29: Display "K" 30: Display "N" 31: Display "O"	1901	-	★
FE-02	Floor 2 display		1902	-	★
FE-03	Floor 3 display		1903	-	★
FE-04	Floor 4 display		1904	-	★
FE-05	Floor 5 display		1905	-	★
FE-06	Floor 6 display		1906	-	★
FE-07	Floor 7 display		1907	-	★
FE-08	Floor 8 display		1908	-	★
FE-09	Floor 9 display		1909	-	★
FE-10	Floor 10 display		0100	-	★
FE-11	Floor 11 display		0101	-	★
FE-12	Floor 12 display		0102	-	★
FE-13 to FE-52	Floor 13 to floor 52 display		0103–0502	-	★
FE-53	Floor 53 display		0503	-	★
FE-54	Floor 54 display		0504	-	★
FE-55	Floor 55 display		0505	-	★

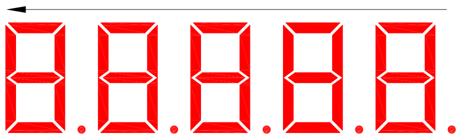
Function Code	Parameter Name	Setting Range	Default	Unit	Property
FE-56	Floor 56 display	32: Display "Q" 33: Display "S" 34: Display "T" 35: Display "U" 36: Display "V" 37: Display "W" 38: Display "X" 39: Display "Y" 40: Display "Z" 41: Display "15" 42: Display "17" 43: Display "19"	0506	-	★
FE-61	Special display 1		0	-	★
FE-62	Special display 2		0	-	★
FE-63	Special display 3		0	-	★
FE-64	Special display 4		0	-	★
FE-65	Special display 5		0	-	★
Group FH: Closed-loop parameter					
FH-00	Security password for group FH	0–65535	01000	-	☆
FH-01	Encoder type	0: SIN/COS encoder, absolute encoder 1: UVW encoder 4: ABZ incremental encoder	0	-	★
FH-02	Encoder pulses per revolution	0–10000	2048	PPR	★
FH-03	Encoder wire-breaking detection time	0–10.0	2.1	s	★
FH-04	Encoder initial angle	0.0–359.9	0	° (Degree)	★
FH-05	Encoder current angle	0.0–359.9	0	° (Degree)	●
FH-06	Wiring mode	0–15	0	-	★
FH-08	Signal zero drift 1	0–65535	0	-	★
FH-09	Signal zero drift 2	0–65535	0	-	★
FH-10	Signal zero drift 3	0–65535	0	-	★
FH-11	Signal zero drift 4	0–65535	0	-	★
FH-17	Position of up slow-down 1	0.000–300.00	0.00	m	★
FH-18	Position of down slow-down 1	0.000–300.00	0.00	m	★
FH-19	Position of up slow-down 2	0.000–300.00	0.00	m	★
FH-20	Position of down slow-down 2	0.000–300.00	0.00	m	★
FH-21	Position of up slow-down 3	0.000–300.00	0.00	m	★
FH-22	Position of down slow-down 3	0.000–300.00	0.00	m	★
FH-29	Position switch delay	0–200	0	ms	★
Group FL: Extended terminal function					

<table>
<tr><th>Function Code</th><th>Parameter Name</th><th>Setting Range</th><th>Default</th><th>Unit</th><th>Property</th></tr>
<tr><td>FL-00</td><td>SSL7020 input type</td><td>0–2048</td><td>1856</td><td>-</td><td>★</td></tr>
<tr><td colspan="6">The NO/NC setting of X1 to X11 on the SSL7020 is described in the following table.
<table>
<tr><th>Bit</th><th>Parameter Name</th><th>Default</th><th>Bit</th><th>Parameter Name</th><th>Default</th></tr>
<tr><td>Bit0</td><td>Door 1 light curtain</td><td>0</td><td>Bit6</td><td>Full-load signal (digital)</td><td>1</td></tr>
<tr><td>Bit1</td><td>Door 2 light curtain</td><td>0</td><td>Bit7</td><td>Overload signal (digital)</td><td>0</td></tr>
<tr><td>Bit2</td><td>Door 1 open limit</td><td>0</td><td>Bit8</td><td>Lightload signal (digital)</td><td>1</td></tr>
<tr><td>Bit3</td><td>Door 2 open limit</td><td>0</td><td>Bit9</td><td>Door machine 1 safety edge signal</td><td>1</td></tr>
<tr><td>Bit4</td><td>Door 1 close limit</td><td>0</td><td>Bit10</td><td>Door machine 2 safety edge signal</td><td>1</td></tr>
<tr><td>Bit5</td><td>Door machine 2 close limit</td><td>0</td><td colspan="3">0: NC input; 1: NO input</td></tr>
</table></td></tr>
<tr><td>FL-01</td><td>SSL7040: JP1 input</td><td rowspan="6">0: Reserved
1: Light load signal
2: Half load signal
3: Door 2 selection
4: Door 2 restricted
5: Door 1 safety edge
6: Door 2 safety edge
7: Single/Double door selection
8: Fire emergency floor switchover</td><td>0</td><td>-</td><td>★</td></tr>
<tr><td>FL-02</td><td>SSL7040: JP2 input</td><td>0</td><td>-</td><td>★</td></tr>
<tr><td>FL-03</td><td>SSL7040: JP3 nput</td><td>0</td><td>-</td><td>★</td></tr>
<tr><td>FL-04</td><td>SSL7040: JP4 nput</td><td>0</td><td>-</td><td>★</td></tr>
<tr><td>FL-05</td><td>SSL7040: JP5 nput</td><td>0</td><td>-</td><td>★</td></tr>
<tr><td>FL-06</td><td>SSL7040: JP5 nput</td><td>0</td><td>-</td><td>★</td></tr>
<tr><td>FL-07</td><td>SSL7040: A1 output</td><td rowspan="10">0: Reserved
1: Fault output
2: Non-door zone stop output
3: Non-service state output
4: Fire emergency output
5: Power failure emergency output
6: Door lock valid
7: Night output signal
8: Fire indicator
9: Fire announcement</td><td>1</td><td>-</td><td>★</td></tr>
<tr><td>FL-08</td><td>SSL7040: A2 output</td><td>2</td><td>-</td><td>★</td></tr>
<tr><td>FL-09</td><td>SSL7040: B1 output</td><td>3</td><td>-</td><td>★</td></tr>
<tr><td>FL-10</td><td>SSL7040: B2 output</td><td>4</td><td>-</td><td>★</td></tr>
<tr><td>FL-11</td><td>SSL7040: C1 output</td><td>5</td><td>-</td><td>★</td></tr>
<tr><td>FL-12</td><td>SSL7040: C2 output</td><td>6</td><td>-</td><td>★</td></tr>
<tr><td>FL-13</td><td>SSL7040: C3 output</td><td>0</td><td>-</td><td>★</td></tr>
<tr><td>FL-14</td><td>SSL7040: C4 output</td><td>7</td><td>-</td><td>★</td></tr>
<tr><td>FL-15</td><td>SSL7040: C5 output</td><td>0</td><td>-</td><td>★</td></tr>
<tr><td>FL-16</td><td>SSL7040: C6 output</td><td>8</td><td>-</td><td>★</td></tr>
<tr><td>FL-17</td><td>SSL7040: JP1 input</td><td rowspan="2">1: Elevator lock signal
2: Fire emergency signal
3: Current floor forbidden
4: VIP floor signal
5: Security floor signal
6: Service floor switchover
7: Parking floor switchover
8: Down collective selective switch
9: Peak service switch
10: Fire emergency floor switchover</td><td>1</td><td>-</td><td>★</td></tr>
<tr><td>FL-18</td><td>SSL7040: JP2 input</td><td>2</td><td>-</td><td>★</td></tr>
<tr><td>FL-19</td><td>SSL7040: JP1 output</td><td rowspan="2">0: Reserved
1: Up arrival indicator
2: Down arrival indicator
3: Fault output
4: Non-door zone stop output
5: Non-service state output
6: Buzzer output</td><td>1</td><td>-</td><td>★</td></tr>
<tr><td>FL-20</td><td>SSL7040: JP2 output</td><td>2</td><td>-</td><td>★</td></tr>
</table>

Function Code	Parameter Name	Setting Range	Default	Unit	Property
Group Fr: Leveling adjustment parameters					
Fr-00	Leveling adjustment function	**0: Disabled** **1: Enabled**	0	-	★
Fr-01	Leveling adjustment record 1	0–60060	30030	-	★
...	...		...	-	
Fr-28	Leveling adjustment record 28		30030	-	★
Group FU: Monitor parameters					
FU-03	Pre-torque current	0.0–200.0	0.0	%	●
FU-04	Logic information	0–65535	0	-	●

The LEDs are arranged as 5, 4, 3, 2, 1 from left to right. LEDs 1, 2 and 3 respectively display the states of door 1, door 2, and car. LEDs 4 and 5 together show the elevator state.

5 4 3 2 1



The LEDs are defined in the following table.

5		4		3		2		1	
Elevator State				Car State		Door 2 state		Door 1 state	
00	Inspection state	08	Elevator lock	0	Waiting state	0	Waiting state	0	Waiting state
01	Shaft auto-tuning	09	Idle elevator parking	1	About to stop	1	Door open state	1	Door open state
02	Micro-leveling	10	Re-leveling at inspection speed	2	Running state	2	Door open limit	2	Door open limit
03	Returning to base floor at fire emergency	11	Emergency evacuation operation			3	Door close state	3	Door close state
04	Firefighter operation	12	Motor auto-tuning			4	Door close limit	4	Door close limit
05	Fault state	13	Keypad control						
06	Attendant operation	14	Base floor verification						
07	Automatic running								

Function Code	Parameter Name	Setting Range	Default	Unit	Property
FU-05	Curve information	0–65535	0	-	●

LEDs 5, 4 and 3 have no display, while LEDs 2 and 1 show the running curve information, as described in the following table.

5	4	3	2		1	
No Display	**No Display**	**No Display**	**Curve Information**			
–	–	–	00	Standby state	08	Stable-speed running segment

Function Code	Parameter Name	Setting Range	Default	Unit	Property

01	Zero-speed start segment	09	Deceleration start segment
02	Zero-speed holding segment	10	Linear deceleration segment
03	Reserved	11	Deceleration end segment
04	Startup speed stage	12	Zero speed at stop
05	Acceleration start segment	13	Current stop stage
06	Linear acceleration segment	14	Reserved
07	Acceleration end segment	15	Stop data processing

Function Code	Parameter Name	Setting Range	Default	Unit	Property
FU-06	Set speed	0.000–8.000	0	m/s	●
FU-07	Feedback speed	0.000–8.000	0	m/s	●
FU-08	Bus voltage	0–999.9	0	V	●
FU-09	Output voltage	0–999.9	0	V	●
FU-10	Output current	0–655.00	0	A	●
FU-11	Output frequency	0.00–99.99	0	Hz	●
FU-12	Output voltage	0.0–200.0	0	%	●
FU-13	Torque current	0–655.00	0	A	●
FU-14	Output power	0.00–99.99	0	kW	●
FU-15	Present position	0.00–300.00	0	m	●
FU-16	Communication interference	0–65535	0	-	●

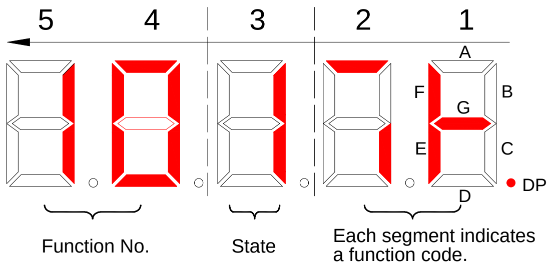
It displays the current communication quality of the system, as described in the following table.

5		4		3		2		1	
SPI Communication		MOD2 **Communication**		CAN2 **Communication**		MOD1 **Communication**		CAN1 **Communication**	
0	Good	0	Good	0	Good	0	Good	0	Good
↓	↑	↓	↑	↓	↑	↓	↑	↓	↑
9	Interrupted	9	Interrupted	9	Interrupted	9	Interrupted	9	Interrupted

Function Code	Parameter Name	Setting Range	Default	Unit	Property
FU-17	Encoder interference	0–65535	0	-	●
FU-18	Input state 1	0–65535	0	-	●
FU-19	Input state 2	0–65535	0	-	●
FU-20	Input state 3	0–65535	0	-	●
FU-22	Input state 5	0–65535	0	-	●
FU-23	Output state 1	0–65535	0	-	●

Function Code	Parameter Name	Setting Range	Default	Unit	Property
FU-24	Output state 2	0–65535	0	-	●
FU-25	Output state 3	0–65535	0	-	●
FU-26	Car input state	0–65535	0	-	●
FU-27	Car output state	0–65535	0	-	●
FU-28	Hall sate	0–65535	0	-	●
FU-29	System state 1	0–65535	0	-	●
FU-30	System state 2	0–65535	0	-	●

The following figure shows an example of the displayed input states.
The LEDs from right to left are numbered 1, 2, 3, 4, and 5. LEDs 5 and 4 show the function No.; LED 3 shows whether the function is valid (1) or invalid (0); the 16 segments of LEDs 1 and 2 show the states of the 16 functions in this parameter.
According to the figure, LEDs 5, 4, and 3 show that function 10 (Inspection down) is 1 (Valid); LEDs 1 and 2 show that besides function 10, functions 4 (Safety circuit feedback), 5 (Door lock circuit feedback), 6 (RUN contactor feedback), 7 (Brake contactor feedback), and 8 (Inspection signal) are valid.



FU-18 **Input state 1**			
LEDs 5 and 4			
00	Reserved	08	Inspection signal
01	Up leveling signal	09	Inspection up
02	Down leveling signal	10	Inspection down
03	Door zone signal	11	Fire emergency signal
04	Safety circuit feedback	12	Up limit signal
05	Door lock circuit feedback	13	Down limit signal
06	RUN contactor feedback	14	Overload signal
07	Brake contactor feedback	15	Full-load signal

FU-19 **Input state 2**			
LEDs 5 and 4			
16	Up slow-down 1 signal	24	Door machine 1 light curtain
17	Down slow-down 1 signal	25	Door machine 2 light curtain
18	Up slow-down 2 signal	26	Brake contactor feedback 2
19	Down slow-down 2 signal	27	UPS input
20	Up slow-down 3 signal	28	Elevator lock input
21	Down slow-down 3 signal	29	Safety circuit 2 signal
22	Shorting door lock circuit contactor feedback	30	Shorting motor stator contactor feedback
23	Firefighter running signal	31	Reserved

FU-20 **Input state 3**			
LEDs 5 and 4			
32	Motor overheat signal	40	Peak service switch
33	VIP signal	41	Fire emergency enable

FU-22 **Input state 5**			
LEDs 5 and 4			
00	Reserved	08	Reserved
01	Reserved	09	Reserved

Function Code	Parameter Name	Setting Range	Default	Unit	Property

34	Earthquake signal	42	Back door selection	02	Reserved	10	Reserved
35	Security signal	43	Back door foridden	03	Reserved	11	Reserved
36	Service floor switchover	44	Light load	04	Higher-voltage safety circuit signal	12	Reserved
37	Fire emergency floor switchover	45	Half load	05	Higher-voltage door lock circuit signal	13	Reserved
38	Parking floor switchover	46	Double door control switch	06	Reserved	14	Reserved
39	Down collective selective switch	47	Motor input	07	Reserved	15	Reserved

FU-23 **Output state 1**				FU-24 **Output state 2**			
LEDs 5 and 4				LEDs 5 and 4			
00	Reserved	08	Door 2 close	16	Higher-voltage startup of brake	24	Reserved
01	RUN contactor output	09	Reserved	17	Elevator running in up direction	25	Reserved
02	Brake contactor output	10	Fault state above level 3	18	Lamp/Fan output	26	Reserved
03	Shorting door lock circuit contactor output	11	Running state	19	Medical sterilization	27	Reserved
04	Fire emergency floor arrival	12	Shorting PMSM stator contactor output	20	Non-door zone stop	28	Reserved
05	Door 1 open	13	Power failure emergency output	21	Electric lock output	29	Reserved
06	Door 1 close	14	Reserved	22	Non-service state	30	Reserved
07	Door 2 open	15	Emergency tweet	23	Reserved	31	Reserved

FU-26 **Car input state**				FU-27 **Car output state**			
LEDs 5 and 4				LEDs 5 and 4			
00	Door 1 light curtain	08	Light-load input	00	Fan/Lamp output	08	Down arrival gong
01	Door 2 light curtain	09	Reserved	01	Door 1 open	09	Reserved
02	Door 1 open limit	10	Reserved	02	Door 1 close	10	Reserved
03	Door 2 open limit	11	Reserved	03	Forced door close 1	11	Reserved
04	Door 1 close limit	12	Reserved	04	Door 2 open	12	Reserved
05	Door 2 close limit	13	Reserved	05	Door 2 close	13	Reserved
06	Full-load input	14	Reserved	06	Forced door close 2	14	Reserved
07	Overload input	15	Reserved	07	Up arrival gong	15	Reserved

FU-28 **Hall state**		FU-29 **System state 1**		FU-30 **System state 2**	
LEDs 5 and 4		LEDs 5 and 4		LEDs 5 and 4	
00	Reserved	00	Door open 1 button	00	Up direction display
01	Elevator lock signal	01	Door close 1 button	01	Down direction display
02	Fire emergency signal	02	Door open delay 1	02	Running state

Function Code	Parameter Name	Setting Range	Default	Unit	Property

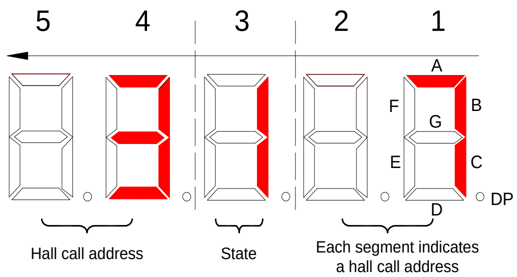
03	Current floor forbidden	03	Direct travel ride switch	03	System full-load
04	VIP signal	04	Attendant switch	04	System overload
05	Security signal	05	Direction change switch	05	System half-load
06	Service floor switchover	06	Independent running switch	06	System light-load
07	Parking floor switchover	07	Fire emergency 2 switch	07	Commissioning enabled
08	Down collective selective switch	08	Door open 2 button	08	Maintenance enabled
09	Peak service switch	09	Door close 2 button	09	Peak service enabled
10	Fire emergency floor switchover	10	Door open delay 2	10	Reserved

Function Code	Parameter Name	Setting Range	Default	Unit	Property
FU-31	Car load	0–255	0	-	●
FU-32	Nearest landing floor	1–56	0	-	●
FU-33	Destination floor	1–56	0	-	●
FU-34	Time to arrive	0.0–60.0	0	s	●
FU-35	Deceleration distance	0.0–100.0	0	m	●
FU-49	Mod1 hall call state 1	0–65535	0	-	●
FU-50	Mod1 hall call state 2	0–65535	0	-	●
FU-51	Mod1 hall call state 3	0–65535	0	-	●
FU-52	Mod1 hall call state 4	0–65535	0	-	●
FU-53	Mod2 hall call state 1	0–65535	0	-	●
FU-54	Mod2 hall call state 2	0–65535	0	-	●
FU-55	Mod2 hall call state 3	0–65535	0	-	●
FU-56	Mod2 hall call state 4	0–65535	0	-	●

These parameters display the communication state between SSL7040s of all floors and the SSL-7000-A1.

State 1, state 2, state 3, and state 4 respectively indicate the communication state of floors 1 to 16, 17 to 31, 32 to 47, and 48 to 56.

As shown in the following figure, LEDs 5 and 4 show the floor address; LED 3 shows whether the communication for this floor address is normal ("1" is displayed) or interrupted ("0" is displayed). The communication quality can be viewed from LEDs 1 and 2: The 16 segments show the communication state of 16 floor addresses; ON indicates that the communication is normal, and OFF indicates that the communication is interrupted.



Group FF: Factory parameters

Function Code	Parameter Name	Setting Range	Default	Unit	Property
Group FP: **User parameters**					
FP-00	User password	0–65535	0	-	☆
FP-01	Parameter update	0: No operation 1: Restore default settings 2: Clear fault records 3: Restore logic board parameters 4: Restore all parameters	0	-	★
FP-02	User-defined parameter display	0: Invalid 1: Valid	0	-	★
Group E0: 1st fault information					
E0-00	1st fault code	0–6299	0	-	●
E0-01	1st fault subcode	0–65535	0	-	●
E0-02	1st fault month and day	0–1231	0	MM.DD	●
E0-03	1st fault month and day	0–2359	0	HH.MM	●
E0-04	Logic information of 1st fault	0–65535	0	-	●
E0-05	Curve information of 1st fault	0–65535	0	-	●
E0-06	Set speed upon 1st fault	0.000–8.000	0	m/s	●
E0-07	Feedback speed upon 1st fault	0.000–8.000	0	m/s	●
E0-08	Bus voltage upon 1st fault	0–999.9	0	V	●
E0-09	Output voltage 1st fault	0–999.9	0	V	●
E0-10	Output current upon 1st fault	0–655.00	0	A	●
E0-11	Output frequency upon 1st fault	0.00–99.99	0	Hz	●
E0-12	Output torque upon 1st faul	0–100	0	%	●
E0-13	Torque current upon 1st fault	0–655.00	0	A	●
E0-14	Output power upon 1st fault	0.00–99.99	0	kW	●
E0-15	Current position upon 1st fault	0.00–300.00	0	m	●
E0-16	Communication interference upon 1st fault	0–65535	0	-	●
E0-17	Encoder interference upon 1st fault	0–65535	0	-	●
E0-18	Input state 1 upon 1st fault	0–65535	0	-	●
E0-19	Input state 2 upon 1st fault	0–65535	0	-	●

Function Code	Parameter Name	Setting Range	Default	Unit	Property
E0-20	Input state 3 upon 1st fault	0–65535	0	-	●
E0-21	Input state 4 upon 1st fault	0–65535	0	-	●
E0-22	Input state 5 upon 1st fault	0–65535	0	-	●
E0-23	Output state 1 upon 1st fault	0–65535	0	-	●
E0-24	Output state 2 upon 1st fault	0–65535	0	-	●
E0-25	Output state 3 upon 1st fault	0–65535	0	-	●
E0-26	Car input state upon 1st fault	0–65535	0	-	●
E0-27	Car output state upon 1st fault	0–65535	0	-	●
E0-28	Hall call state upon 1st fault	0–65535	0	-	●
E0-29	System state 1 upon 1st fault	0–65535	0	-	●
E0-30	System state 2 upon 1st fault	0–65535	0	-	●
E0-31	Car load upon 1st fault	0–255	0	-	●
Groups E1 to E9 The parameters of groups E1 to E9 are similar to those of group E0, and define the information of the subsequent 9 faults.					

Chapter 4 System Application

4.1 Parallel or Group Control

4.1.1 Parallel Control of Two Elevators

The SLEC7000 supports parallel/group control of a maximum of 8 elevators implemented at CAN2.

The following table lists the parameters to be set for parallel control.

Table 4-1 Parameter setting of parallel control

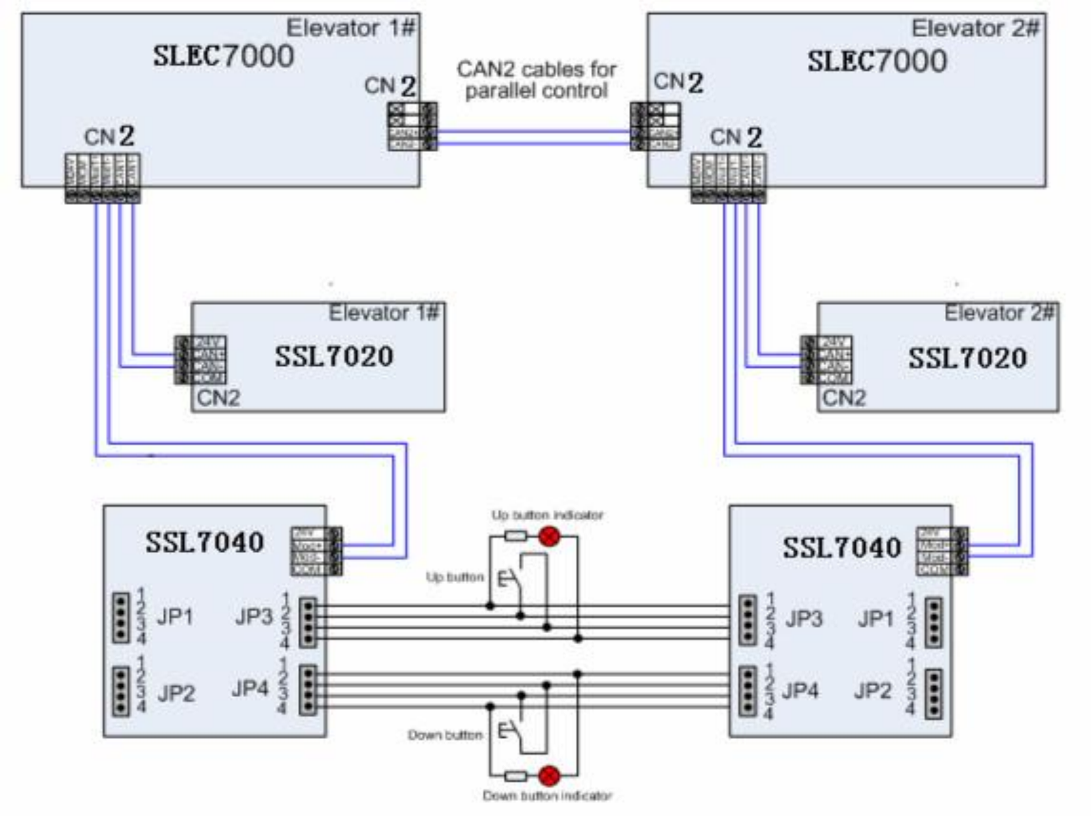
Function Code	Parameter Name	Setting Range	Setting in Parallel Control
Fd-00	Number of elevators in parallel/group mode	1–8	2
Fd-01	Elevator No.	1–8	Master elevator: 1
			Slave elevator: 2
Fd-02	Group control mode	0: Parallel control	Applicable to parallel control of 2 elevators
		1:Group control of destination dispatch	Applicable to group control of destination dispatch
		2: Group control of traditional mode	Applicable to group control of common up/down collective selective (non-destination dispatch group control)

Note

When the CAN2 communication port is used for parallel control, you need not set the SSL7020 address switch.

The following figure shows the wiring for parallel control.

Figure 4-1 Wiring for parallel control



4.1.2 Address Setting of Physical Floors

Physical floor, relative to the SLEC7000 control system, is defined by the installation position of the leveling plate. The floor (such as the ground floor) at which the lowest leveling plate is installed corresponds to physical floor 1. The top physical floor is the accumulative number of the leveling plates. In parallel control mode, the physical floor numbers of the same floor for two elevators are consistent.

If the floor structures of two elevators are different, physical floors should start with the floor with the lowest position. The physical floors at the overlapped area of the two elevators are the same. Even if one elevator does not stop a floor in the overlapped area, a leveling plate should be installed there. You can make the elevator not stop at the floor by setting service floors.

When two elevators are in parallel mode, the addresses of the SSL7040s should be set according to physical floors. Parallel running can be implemented only when the SSL7040 address set for one elevator is the same as that for the other elevator in terms of the same floor.

Note

In parallel mode, the top floor (F6-00) and bottom floor (F6-01) of the elevators should be set based on corresponding physical floors.

Assume that there are two elevators in parallel mode. Elevator 1 stops at floor B1, floor 1, floor 2, and floor 3, while elevator 2 stops at floor 1, floor 3, and floor 4. Now, you need to set related parameters and SSL7040 addresses according to the following table.

Table 4-2 Parameter setting and SSL7040 addresses of two elevators

	Elevator 1	Elevator 2
Number of elevators in parallel/group mode (Fd-00)	2	2
Elevator No. (Fd-01)	1	2

Actual floor	Physical floor	SSL7040 address	SSL7040 display	SSL7040 address	SSL7040 display
B1	1	1	FE-01 = 1101		
1	2	2	FE-02 = 1901	2	FE-02 = 1901
2	3	3	FE-03 = 1902	Non-stop floor but leveling plate required	FE-03 = 1902
3	4	4	FE-04 = 1903	4	FE-04 = 1903
4	5			5	FE-05 = 1904
Bottom floor (F6-01)		1		2	
Top floor (F6-00)		4		5	
Service floor (F6-05)		65535		65531 (not stop at physical floor 3)	

4.1.3 Group Control

For the use of group control of three and more elevators, contact Monarch.

4.2 Emergency Evacuation at Power Failure

Passengers may be trapped in the car if power failure suddenly happens during the use of the elevator. The emergency evacuation function at power failure is designed to solve the problem.

The emergency evacuation function is implemented in the following two modes:

- Uninterrupted power supply (UPS)
- Emergency automatic rescue device (ARD) power supply
- Shorting PMSM stator

The three modes are described in detail in the following part.

4.2.1 Shorting PMSM Stator

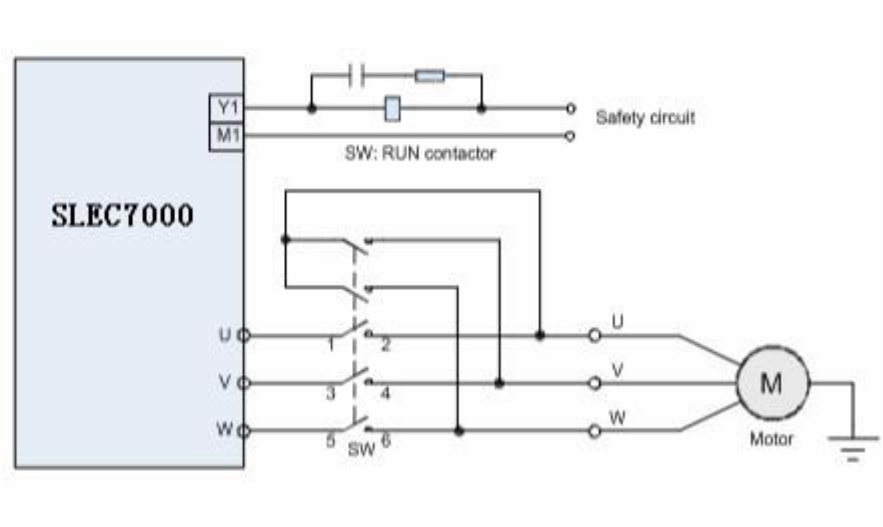
Shorting PMSM stator means shorting phases UVW of the PMSM, which produces resistance to restrict movement of the elevator car. In field application, an auxiliary NC contact is usually added to the NO contact of the output contactor to short PMSM UVW phases to achieve the effect. It is feasible in theory but may cause over-current actually. Due to the poor quality of the contactor and the wiring of adding the auxiliary contact, the residual current of the controller is still high when the outputs UVW are shorted at abnormal stop. This results in an over-current fault and may damage the controller or motor.

We recommend two schemes of shorting PMSM stator.

1) Using integrated shorting PMSM stator contactor

This contactor provides the shorting PMSM stator function itself, featuring safety, reliability and easy wiring. The following figure shows the wiring diagram.

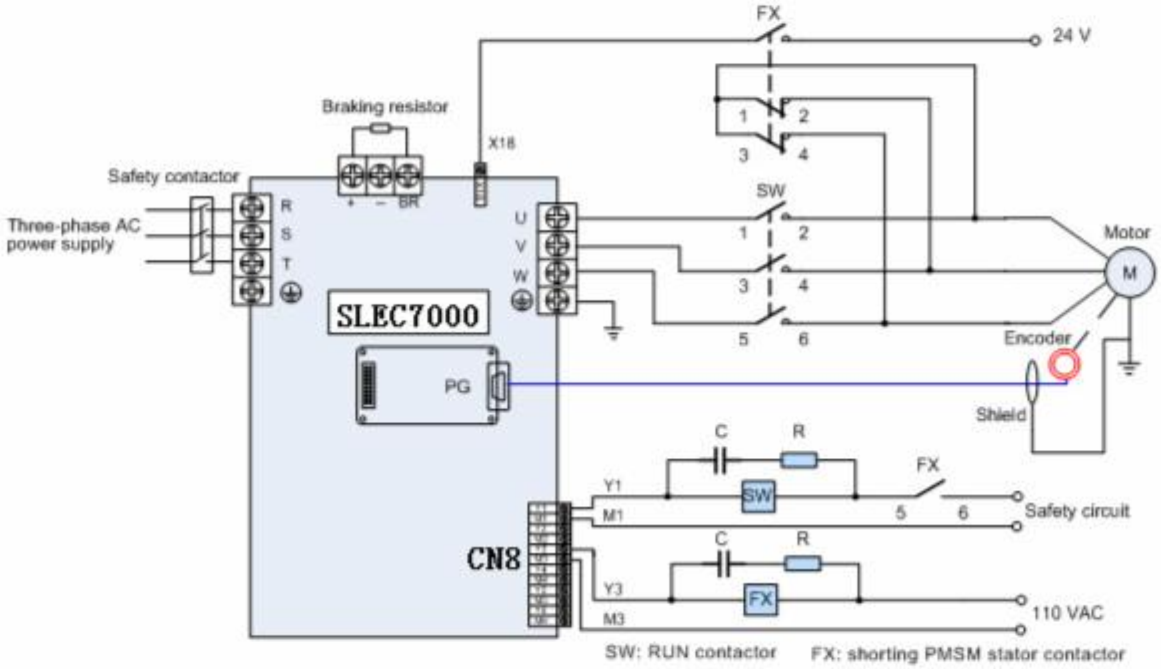
Figure 4-4 Wiring of integrated shorting PMSM stator contactor



2) Using independent shorting PMSM stator contactor

An independent shorting PMSM stator contactor is installed, and a relay NC contactor is used to implement the shorting PMSM stator function. On the coil circuit of the RUN contactor, an NO contact of the shorting PMSM stator contactor is connected in serial, to ensure that output short-circuit does not occur when the parameter setting is incorrect.

Figure 4-5 Wiring of independent shorting PMSM stator contactor



You need to set the shorting PMSM stator contactor (Y4 in this figure); set F5-34 (Y4 function selection) to 12 (Shorting PMSM stator contactor). Set Bit8 of F6-10 based on the contact NO/NC feature: Bit8 = 1: NC output; Bit8 = 0: NO output.

Connect an input terminal to X input to monitor the working status of the contactor, and set the function for this terminal as 30 (NO ouput). In this figure, X14 is used to monitor the working status of the contactor, and therefore, set F5-14 to 30.

4.2.3 Related Configuration

Table 4-3 **Recommended UPS power for each power rating**

UPS Power	Controller Power
1 kVA (700–800 W)	P ≤ 5.5 kW
2 kVA (1400–1600 W)	5.5 kW < P ≤ 11 kW
3 kVA (2100–2400 W)	15 kW

Table 4-4 **Parameter setting under the 220 V UPS scheme**

Function Code	Parameter Name	Setting Range
F3-12	Low-speed self-rescue speed	By default
F3-15	Emergency evacuation speed	By default
F3-16	Emergency evacuation acceleration rate	By default
F5-20 (X20)	Emergency evacuation running	127
F5-37 (Y6)	Y6 function selection	13 (Emergency evacuation automatic switchover)
F6-26	Emergency evacuation lasting time	Default

Table 4-5 Parameter setting of F3-14

<table>
<tr><th>Bit</th><th>Function Description</th><th colspan="6">Setting</th><th>Remarks</th></tr>
<tr><td rowspan="2">Bit0</td><td rowspan="2">Emergency evacuation</td><td>1</td><td colspan="5">Shorting stator braking mode</td><td>-</td></tr>
<tr><td>0</td><td colspan="5">Emergency power supply</td><td>-</td></tr>
<tr><td rowspan="2">Bit1</td><td rowspan="2">Low-voltage drive</td><td>1</td><td colspan="5">48 VDC</td><td>-</td></tr>
<tr><td>0</td><td colspan="5">220 VAC and above</td><td>-</td></tr>
<tr><td rowspan="2">Bit2</td><td rowspan="2">High speed emergnecy evacuation</td><td>1</td><td colspan="5">Allowed</td><td rowspan="2">This function is used when the UPS capacity is sufficient and the voltage is high enough.</td></tr>
<tr><td>0</td><td colspan="5">Forbidden</td></tr>
<tr><td rowspan="2">Bit3</td><td rowspan="2">Door open at single leveling signal</td><td>1</td><td colspan="5">Allowed</td><td rowspan="2">The door can open when a single leveling signal is active.</td></tr>
<tr><td>0</td><td colspan="5">Forbidden</td></tr>
<tr><td>Bit4</td><td rowspan="2">Direction determine mode</td><td>0</td><td rowspan="2">Automatically calculating the direction</td><td>0</td><td rowspan="2">direction determining based on half-load signal)</td><td>1</td><td rowspan="2">Direction of nearest landing floor</td><td>-</td></tr>
<tr><td>Bit5</td><td>0</td><td>1</td><td>0</td><td>-</td></tr>
<tr><td rowspan="2">Bit6</td><td rowspan="2">Stop position</td><td>1</td><td colspan="5">Stop at the base floor</td><td>-</td></tr>
<tr><td>0</td><td colspan="5">Stop at nearest landing floor</td><td>-</td></tr>
<tr><td rowspan="2">Bit7</td><td rowspan="2">Startup compensation</td><td>1</td><td colspan="5">Enabled</td><td>-</td></tr>
<tr><td>0</td><td colspan="5">Disabled</td><td>-</td></tr>
</table>

4.3 Opposite Door Control

The SLEC7000 provides two Modbus communction ports. When double doors are used, communication for door 1 hall calls is implemented by Mod1, and communication for door 2 hall calls is implmented by Mod2. The system supports opposite door control of a maximum of 56 floors.

1. Control modes and related parameter setting

The following table describes the control modes and related parameter setting.

Table 4-6 Control mode and parameter setting

Opposite Door Control Mode		Parameter Setting	Function Description
Mode 1	Simultaneous control	FB-00 = 2 FB-01 = 0	The front door and back door acts simultaneously upon arrival for hall calls and car calls.
Mode 2	Hall call independent, car call simultaneous	FB-00 = 2 FB-01 = 5 (Bit0/Bit2 = 1)	The corresponding door opens upon arrival for hall calls from this door. The front door and back door act simultaneously upon arrival for car calls.
Mode 3	Hall call independent, car call manual control)	FB-00 = 2 FB-01 = 325 (Bit0/Bit2/Bit6/Bit8 = 1)	The corresponding door opens upon arrival for halls call from this door. Upon arrival for car calls, the door to open is selected between the front door and back door by using the door switchover switch. Note that the switchover switch (input function 42/142: Back door selection) can only be connected to the SSL7040 (car I/O extension board) or SSL-7000-A1.
Mode 4	Hall call independent, car call independent	FB-00 = 2 FB-01 = 271 (Bit0/Bit1/Bit2/Bit3/Bit8 = 1)	The corresponding door opens upon arrival for halls call and car calls from this door.

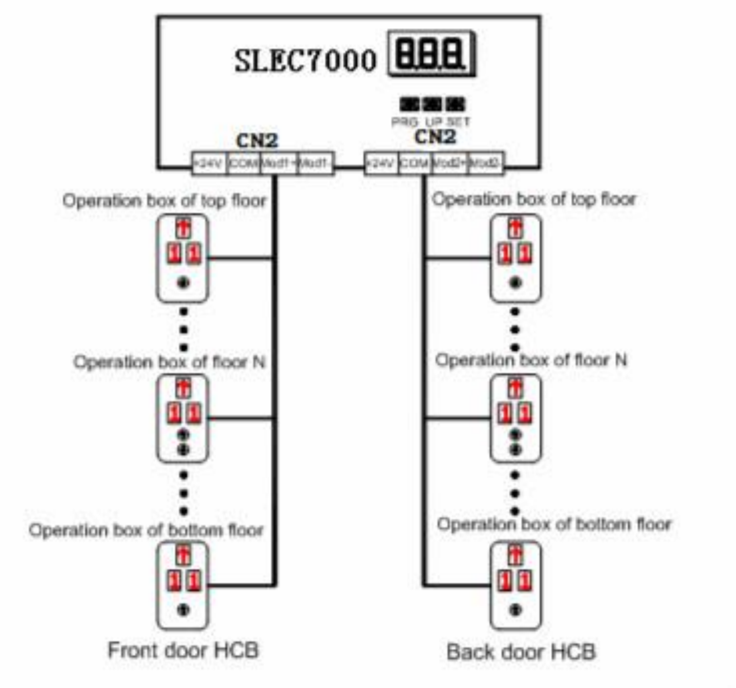
Note

In the fire emergency and elevator lock state, the opposite door is under simultaneous control rather than independent control.

2. Wiring for opposite door control

The hall call addresses of door 1 are consistent with those of door 2, both set based on the actual physical addresses. The following figure shows the wiring diagram.

Figure 4-6 Wiring for opposite door control



Note

Front door and back door are respectively connected to CN7 and CN8 on the SSL7020.

The following table describes the setting of FB-01 for the opposite door function.

Table 4-7 Setting of FB-01 for the opposite door function

Bit	Meaning	Setting		Bit	Meaning	Setting	
Bit0	Hall call button display	1	Front door and back door independent control	Bit1	Car call button display	1	Front door and back door independent control
		0	Front door and back door simultaneous control			0	Front door and back door simultaneous control
Bit2	Hall call door open mode	1	Corresponding door open	Bit3	Car call door open mode	1	Corresponding door open
		0	Both doors open			0	Both doors open
Bit8	Door open button display	1	Front door and back door independent control	Bit9	Door close button display	1	Front door and back door independent control
		0	Front door and back door simultaneous control			0	Front door and back door simultaneous control
Bit10	Door open by door open button	1	Corresponding door open	Bit11	Door close by door close button	1	Corresponding door close
		0	Both doors open			0	Both doors close
Bit6	Front door and back door switchover	1	By switchover switch	Bit13	Doors mutual exclusion selection	1	Always only one door open
		0	Switchover switch invalid			0	Both doors open
Bit14	Light curtain judgment	1	Front door and back door independent control				
		0	Front door and back door simultaneous control				

4.4 Leveling Adjustment

The leveling adjustment method is as follows:

1. Ensure that shaft auto-tuning is completed successfully, and the elevator runs properly at normal speed.
2. Set Fr-00 to 1 to enable the car leveling adjustment function. Then, the elevator shields hall calls, automatically runs to the top floor, and keeps the door open after arrival. If the elevator is at the top floor, it directly keeps the door open.
3. Go into the car, press the top floor button, and the leveling position is changed 1 mm upward; press the bottom floor button, and the leveling position is changed 1 mm downward. The value is displayed in the car.

   Positive value: up arrow + value, negative value: down arrow + value, adjustment range: ±30 mm
4. After completing adjustment for the current floor, press the top floor button and bottom floor button in the car at the same time to save the adjustment result. The car display restores to the normal state. If the leveling position of the current floor need not be adjusted, press the top floor button and bottom floor button in the car at the same time to exit the leveling adjustment state. Then, car calls can be registered.
5. Press the door close button, and press the button for the next floor. The elevator runs to the next floor and keeps the door open after arrival. Then, you can perform leveling adjustment.
6. After completing adjustment for all floors, set Fr-00 to 0 to disable the leveling adjustment function. Otherwise, the elevator cannot be used.

Pay attention to the following precautions during the operation:

1. Each time shaft auto-tuning is performed, all leveling adjustment parameters can be cleared or reserved.

   a. If you set F7-26 to 1 on the operation panel or F7 to 1 on the keypad, all leveling adjustment parameters are reserved.

b. If you set F7-26 to 2 on the operation panel or F7 to 2 on the keypad, all leveling adjustment parameters are reserved.

2. When the re-leveling function is used, the leveling adjustment function is automatically shielded and cannot be used.

3. You need to set Bit6 of FA-15 to 1 to allow leveling adjustment.

4.5 Re-leveling /UCMP

1)Summary

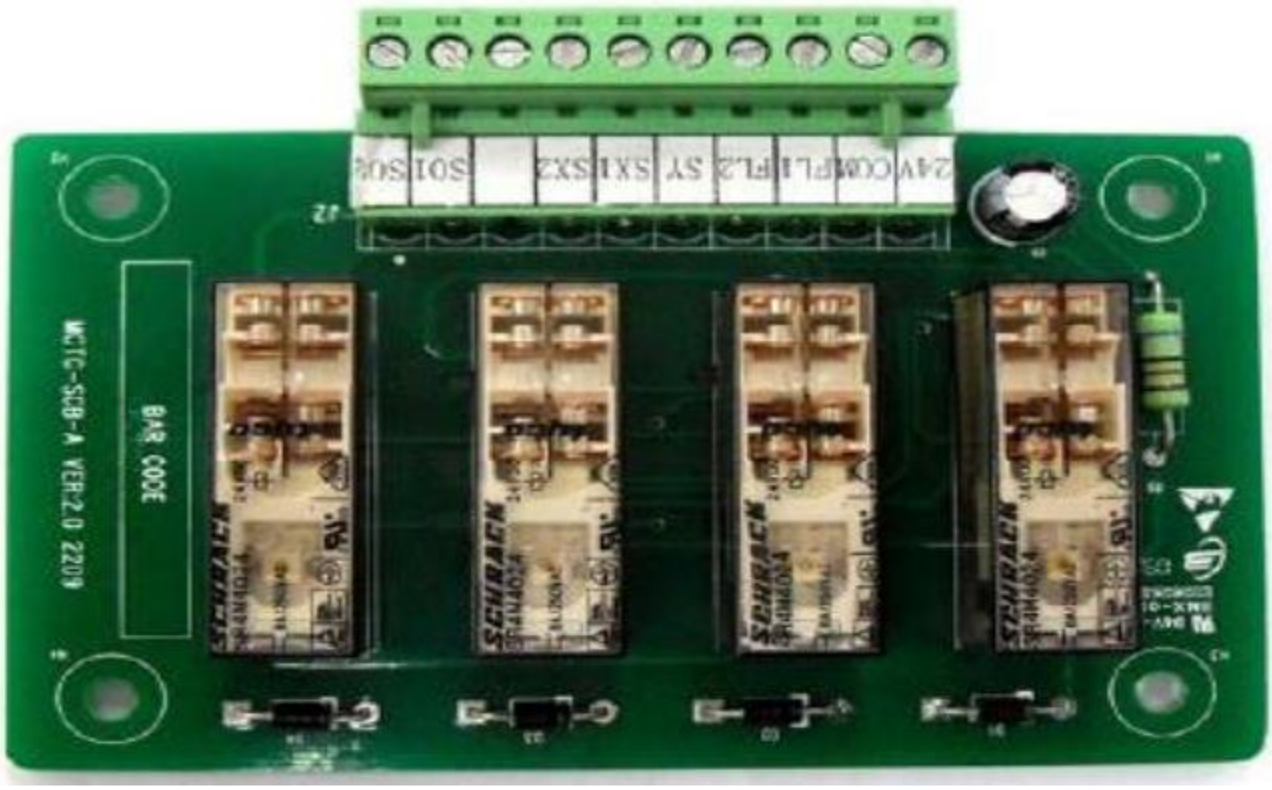
The re-leveling board(door pre-open)is one of the matching product in the SLEC7000 elevator integrated controller system.In conjunction with the SLEC7000 series controller, the re-leveling board (door pre-open) can complete the opening and re-leveling and realize the function of door pre-open.

Re-leveling function: the elevator stops in station, due to the elastic deformation of wire rope or other factors caused by level fluctuations, for people and goods in and out with the inconvenience, the configuration of the re-leveling board (door pre-open) system allows the state to relevel speed automatic running to the floor position.

Door pre-open function: when the elevator in the automatic operation of stopping the process speed is less than 0.3m/s, and at this time the door zone signal effectively, then re-leveling board (door pre-open)through the door lock shorted contactor signal, realize opened in advance, improve the operation efficiency of elevator.

UCMP：The door is not locked in the leveling zone and the car door is not closed, because the driving machine depends on the safe operation of the car accident caused by movement of the car and left stop failure of any single element of the driving control system, the mobile or mobile device to prevent the elevator stop.

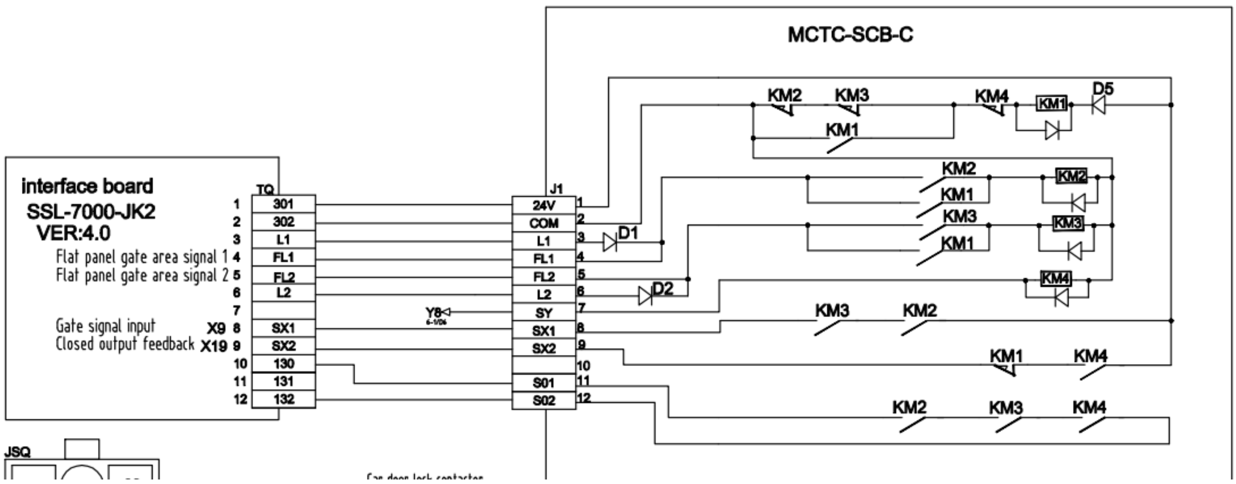
2)picture



Terminal description

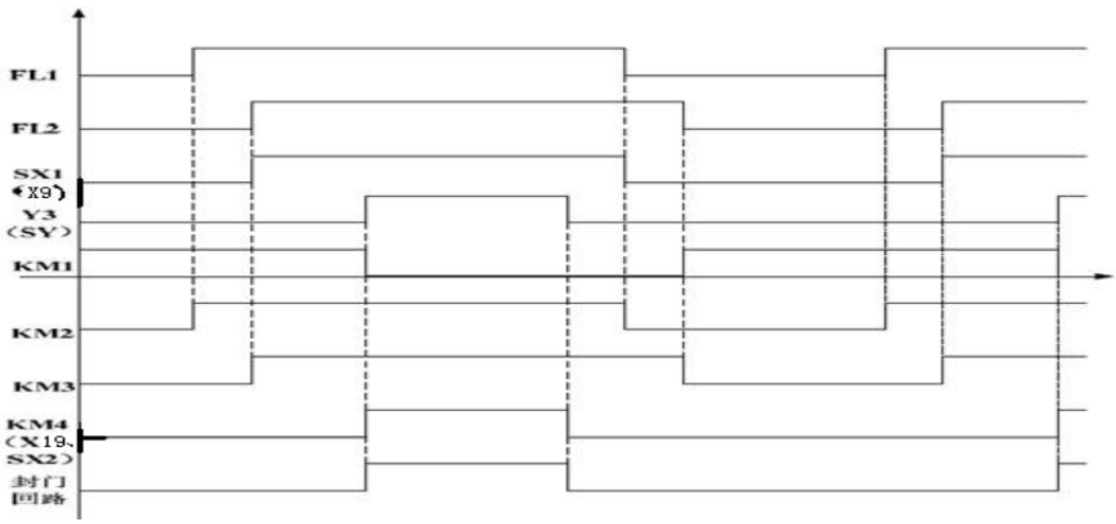
Name	Terminal description
FL1	Re-leveling door zone signal 1
FL2	Re-leveling door zone signal 2
SY	Shorting door lock circuit contactor control output
SX1	Door zone output
SX2	Shorting door lock circuit contactor feedback
SO1,SO2	Door lock circuit

3)wiring diagram



3)Consecution

door pre-open module timing diagram describes the relationship between the relay and the kinds signal , The high level indicates that the signal is valid.

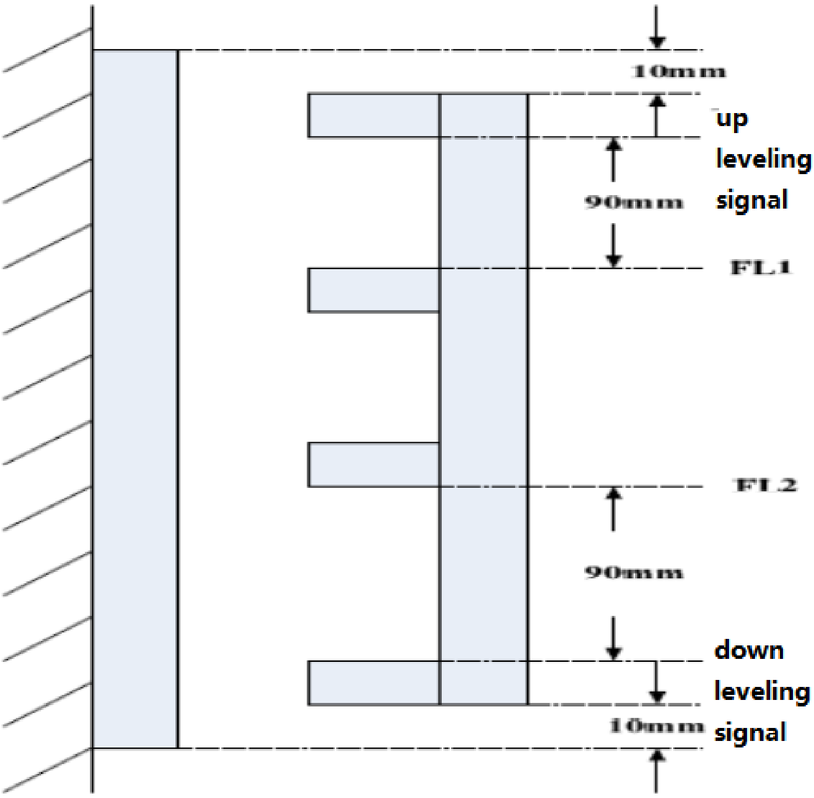


When starting to work on the electrical wiring diagram, according to KM1, the relay will work, the corresponding contact action; when the elevator operation and detection to re-leveling signal 1 (FL1) effective time relay km2, the corresponding contact action; then detected under re-leveling signal 2 (FL2) effective KM3, relay work, the corresponding contact action, also will cause the door zone signal input SX1 (x9), when the slec7000 system detects the signal, open the door ahead output relay Y8 (SY) will be output in the relay km2, KM3 effective and relay km4 effectively, and the relevant contact action, slec7000 detected ahead of the door signal SX2 (X19), and closed the door locks, short locks, ahead of schedule to achieve the effect to open the door. Open the door ahead after the implementation effect, open the door ahead output relay disconnects relay, km4 will not work, the relevant contact action, open the door ahead The input signal is invalid and the sealing lock is broken; when the 1 relevel signal (FL1) is invalid, km2 does not work, and the gate area of input signal is invalid; current relevel signal 2 (FL2) is invalid, the relay KM3 does not work, then the KM1 will work and the corresponding relay contact action.

5)the installation method of the re-leveling sensor and the door zone sensor

Use the door pre-open / re-leveling function need to install leveling sensor. the up leveling sensor, the up re-leveling door zone sensor FL1, and the down leveling sensor ,the down re-leveling door zone sensor FL2, Lower level sensor. The door zone must be installed in order, or re-leveling operation or the door pre-open in advance when the direction will be reversed.

In the field use, if only one door zone sensor signal, please short FL1, FL2, that is, up and down leveling sensor signal as a signal.



6)Description parameter setting

There is re-leveling function, the parameters are set as follows：

Function code	Name	Setting range		default	unit	property
F6-09：Bit2	Re-leveling function	0:invalid 1:re-leveling function		1		
F5-30	X30	0:invalid 1:door lock circuit signal 7:short door lock checking signal		7		

There is door pre-open function, the parameters are set as follows：

Function code	Name	Setting range		default	unit	property
F6-09：Bit3	Re-leveling function	0:invalid 1:door pre-open function		1		
F5-30	X30	0:invalid 1:door lock circuit signal 7:short door lock checking signal		7		

In the actual use of the site, there will be many different situations.The following are several possible scenarios are described.

Situation 1：there is 4 sensor signals,Respectively as up leveling signal,up re-leveling signal input,down re-leveling signal input,down leveling signal.现场 4 个感应器信号，分别为上平层信号，上再平层门区信号输入（FL1），下再平层门区信号输入（FL2），下平层信号。
Processing method：the up and down leveling signals contact to X7,X8 on the main board，The signal of up re-leveling input to FL1 上再平层门区信号输入到 FL1，The signal of down re-leveling input to FL2,all set NO,下再平层门区信号输入到 FL2，均为常开设置，If the sensor set NC 但是如果现场的感应器为常闭设置,那么请使用中间继电器将其转接为常开输入。电梯上行时，当检测到上平层信号、 FL1 信号和 FL2 信号时，主控板对信号进行处理，实现提前开门功能；电梯下行时同理。
相应功能码设置：

Function code	Name	Setting range		default	unit	property
F5-09	X9	000:invalid 003:door zone signal		3		
F5-19：Bit3	X19	000:invalid 022:door pre-open feedback		1		
F5-39	Y8	0:invalid 3:shorting door lock contactor output		3		

In combination with a SLEC7000 system, in accordance with the instructions of F5-09 = 03 normally open door zone signal input, F5-19 =22 door pre-open board normally open feedback input, F5-39 = 3 is shorting door contactor output.

Situation 2: field 3 sensor signals, respectively, the upper level signal, the level zone signal, the lower level signal

Processing method: the upper and lower level signal is connected with the main control board of X7 and X8, the signal of the broken door area is connected with the X9 of the main control board, and the signal of the door area is connected to the FL1 and the FL2, that is to say, the FL1 and the FL2 simultaneously connect the signal of the gate area. If the gate area signal is normally closed, use the relay to switch to normally open input. When the elevator up, detection of the upper level signal, and then detect the FL1 and FL2 at the same time, when the effective opening of the door function; down the same line. The corresponding function code settings: same as case 1

Please note: 2, with FL1 and FL2 test (professionals to monitor this). From the security point of view, we recommend using 2 flat gate signal to ensure the safe and effective operation of the system.

7) UCMP testing :

no-load (full-load) car test procedure

①make sure the car in no-load(full-load)state

②all floors hall call shielding

③operate the elevator to stop in door section normally

④Turn the controller button to “emergency electric running state” after door normally closed.

⑤F-8(of control system function small keyboard) =7, if the digital pipe shows E88 frequently, the UCMP function is opened.

⑥disconnect the electric circuit of door lock by extract pin JK1 (UCTS) (landing door, car door mechanical structure should still remain closed).

⑦hold inspection up run (down run) button when in no-load (full-load) state, door sealed contactor output, door lock short connected, and the elevator runs with inspection speed.

⑧After leaving door section, the hardware UCMP will cancel door lock short connection, and the elevator shows E65 (UCMP fault), the elevator stops within speed under international standard, the UCMP test is succeed. Restore the JK1(UCTS) after completion.

⑨Press RES button of operator, cancel E65 fault, the elevator turns to normal condition.

4.6 Brake force test

1. manual test steps:

① elevator runs normally;

② set F-8=8 by control system function code small keyboard, car call, hall call, door opening should be shielded;

③ door lock is effective, star contactor output, operating contactor output;

④ DSP run instruction is effective, DSP output torque test the brake force;

⑤ DSP test completed, the control system closed star contactor output, operating contactor output;

⑥ result shows in F7-15 of handhold operator and digital pipe of main board, 1 indicates qualified, 2 unqualified with E66 fault, retest is needed after adjustment;

⑦Return the hall car, door function.

2. auto test steps:

① elevator runs normally;

② the auto test function will be open while the F7-17 is not 0, for example: F7-17-60, the auto test period is 60 days;

③ actual test time should be 5 more minutes of F7-18

Function code	Function declaration	Setting range	factory	remarks
F7-15	result	0~2	0	1：qualified 2：unqualified
F7-16	rest time	0~60	0	/
F7-17	period	0~60	60	/0 no brake test function
F7-18	test time	00:00~23:59	0	/

④the elevator stops automatically if the period time is on, and over the set time of F7-18;

⑤ the elevator enters auto brake force monitoring, shielding car, hall call, door open with 15 mins no out calling;

⑥ door lock is effective, star contactor output, operating contactor output;

⑦ DSP run instruction is effective, DSP output torque test the brake force;

⑧ DSP test completed, the control system closed star contactor output, operating contactor output;

⑨ result shows in F7-15 of handhold operator and digital pipe of main board, 1 indicates qualified, 2 unqualified with E66 fault, retest is needed after adjustment;

⑩ Return the hall car, door function.

Chapter 5 Troubleshooting

5.1 Description of Fault Levels

The SLEC7000 has almost 60 pieces of alarm information and protective functions. It monitors various input signals, running conditions and feedback signals. If a fault occurs, the system implements the relevant protective function and displays the fault code.

The controller is a complicated electronic control system and the displayed fault information is graded into five levels according to the severity. The faults of different levels are handled according to the following table.

Table 5-1 Fault levels

Category	Action	Remarks
Level 1	1. Display the fault code. 2. Output the fault relay action command.	1A. The elevator running is not affected on any condition.
Level 2	1. Display fault code. 2. Output the fault relay action command. 3. Continue normal running of the elevator.	2A. The parallel/group control function is disabled.
		2B. The door pre-open/re-leveling function is disabled.
Level 3	1. Display the fault code. 2. Output the fault relay action command. 3. Stop output and apply the brake immediately after stop.	3A. In low-speed running, the elevator stops at special deceleration rate, and cannot restart.
		3B. In low-speed running, the elevator does not stop. In normal-speed running, the elevator stops, and then can start running at low speed after a delay of 3s.
Level 4	1. Display the fault code. 2. Output the fault relay action command. 3. In distance control, the elevator decelerates to stop and cannot run again.	4A. In low-speed running, the elevator stops under special deceleration rate, and cannot restart.
		4B. In low-speed running, the elevator does not stop. In normal-speed running, the elevator stops, and then can start running at low speed after a delay of 3s.
		4C. In low-speed running, the elevator does not stop. In normal-speed running, the elevator stops, and then can start running at low speed after a delay of 3s.
Level 5	1. Display the fault code. 2. Output the fault relay action command. 3. The elevator stops immediately.	5A. In low-speed running, the elevator stops immediately and cannot restart.
		5B. In low-speed running, the elevator does not stop. In normal-speed running, the elevator stops, and then can start running at low speed after a delay of 3s.

Note

- A, B, and C are fault sub-category.
- Low-speed running involves inspection, emergency evacuation, shaft auto-tuning, re-leveling, motor auto-tuning, base floor detection, and running in operation panel control.
- Normal-speed running involves automatic running, returning to base floor in fire emergency state, firefighter operation, attendant operation, elevator lock, and elevator parking.

5.2 Fault Information and Troubleshooting

If an alarm is reported, the system performs corresponding processing based on the fault level. You can handle the fault according to the possible causes described in the following table.

Table 5-2 Fault codes and troubleshooting

<table>
<tr><th>Fault Code</th><th>Name</th><th>Possible Causes</th><th>Solution</th><th>Level</th></tr>
<tr><td>E02</td><td>Over-current during acceleration</td><td>● The main circuit output is grounded or short circuited.
● Motor auto-tuning is performed improperly.
● The load is too heavy.
● The encoder signal is incorrect.
● The UPS running feedback signal is incorrect.</td><td rowspan="3">● Check whether the RUN contactor at the controller output side is normal.
● Check:
– Whether the power cable jacket is damaged
– Whether the power cable is possibly short circuited to ground
– Whether the power cable is connected reliably
● Check the insulation of motor power terminals, and check whether the motor winding is short-circuited or grounded.
● Check whether shorting PMSM stator causes controller output short circuit.
● Check whether motor parameters comply with the nameplate.
● Perform motor auto-tuning again.
● Check whether the brake keeps released before the fault occurs and whether the brake is stuck mechanically.
● Check whether the balance coefficient is correct.
● Check whether the encoder wirings are correct. For asynchronous motor, perform SVC and compare the current to judge whether the encoder works properly.
● Check:
– Whether encoder pulses per revolution (PPR) is set correctly
– Whether the encoder signal is interfered with
– Whether the encoder cable runs through the duct independently
– Whether the cable is too long
– Whether the shield is grounded at one end
● Check
– Whether the encoder is installed reliably
– Whether the rotating shaft is connected to the motor shaft reliably
– Whether the encoder is stable during normal-speed running
● Check whether UPS feedback is valid in the non-UPS running state (E02).
● Check whether the acceleration/deceleration rate is too high (E02, E03).</td><td>5A</td></tr>
<tr><td>E03</td><td>Over-current during deceleration</td><td>● The main circuit output is grounded or short circuited.
● Motor auto-tuning is performed improperly.
● The load is too heavy.
● The deceleration rate is too short.
● The encoder signal is incorrect.</td><td>5A</td></tr>
<tr><td>E04</td><td>Over-current at constant speed</td><td>● The main circuit output is grounded or short circuited.
● Motor auto-tuning is performed properly.
● The load is too heavy.
● The encoder is seriously interfered with.</td><td>5A</td></tr>
</table>

Fault Code	Name	Possible Causes	Solution	Level
E05	Over-voltage during acceleration	● The input voltage is too high. ● The regeneration power of the motor is too high. ● The braking resistance is too large, or the braking unit fails. ● The acceleration rate is too short.	● Adjust the input voltage. Observe whether the bus voltage is normal and whether it rises too quickly during running. ● Check for the balance coefficient. ● Select a proper braking resistor and check whether the resistance is too large based on the recommended braking resistance table in chapter 1. ● Check: – Whether the cable connecting the braking resistor is damaged – Whether the cooper wire touches the ground – Whether the connection is reliable	5A
E06	Over-voltage during deceleration	● The input voltage is too high. ● The braking resistance is too large, or the braking unit fails. ● The deceleration rate is too short.		5A
E07	Over-voltage at constant speed	● The input voltage is too high. ● The braking resistance is too large, or the braking unit fails.		5A
E09	Under-voltage	● Instantaneous power failure occurs on the input power supply. ● The input voltage is too low. ● The drive control board fails.	● Eliminate external power supply faults and check whether the power fails during running. ● Check whether the wiring of all power input cables is secure. ● Contact the agent or Monarch.	5A
E10	Controller overload	● The brake circuit is abnormal. ● The load is too heavy. ● The encoder feedback signal is abnormal. ● The motor parameters are incorrect. ● A fault occurs on the motor power cables.	● Check the brake circuit and power input. ● Reduce the load. ● Check whether the encoder feedback signal and setting are correct, and whether the initial angle of the encoder for the PMSM is correct. ● Check the motor parameter setting and perform motor auto-tuning. ● Check the power cables of the motor (refer to the solution of E02).	4A
E11	Motor overload	● FC-02 is set improperly. ● The brake circuit is abnormal. ● The load is too heavy.	● Adjust the parameter (FC-02 can be set to the default value). ● Refer to the solution of E10.	3A
E12	Power supply phase loss	● The power input phases are not symmetric. ● The drive control board fails.	● Check whether the three phases of power supply are balanced and whether the power voltage is normal. If not, adjust the power input. ● Contact the agent or Monarch.	4A
E13	Power output phase loss	● The output wiring of the main circuit is loose. ● The motor is damaged.	● Check the wiring. ● Check whether the contactor on the output side is normal. ● Eliminate the motor fault.	4A

Fault Code	Name	Possible Causes	Solution	Level
E14	Module overheat	● The ambient temperature is too high. ● The fan is damaged. ● The air filter is blocked.	● Lower the ambient temperature. ● Clear the air filter. ● Replace the damaged fan. ● Check whether the installation clearance of the controller satisfies the requirement.	5A
E16	Current control fault	● The excitation current deviation is too large. ● The torque current deviation is too large. ● The torque limit is exceeded for a very long time.	● Check the circuit of the encoder. ● The output MCCB becomes OFF. ● The values of the current loop parameters are too small. ● Perform motor auto-tuning again if the zero-point position is incorrect. ● Reduce the load if it is too heavy.	5A
E17	Reference signal of the encoder incorrect	● The deviation between the Z signal position and the absolute position is too large. ● The deviation between the absolute position angle and the accumulative angle is too large.	● Check whether the encoder runs properly. ● Check whether the encoder wiring is correct and reliable. ● Check whether the PG card wiring is correct. ● Check whether the grounding of the control cabinet and the motor is normal.	5A
E18	Current detection fault	The drive control board fails.	Contact the agent or Monarch.	5A
E19	Motor auto-tuning fault	● The motor cannot rotate properly. ● The motor auto-tuning times out. ● The encoder for the PMSM fails.	● Enter the motor parameters correctly. ● Check the motor wiring and whether phase loss occurs on the contactor at the output side. ● Check the encoder wiring and ensure that the encoder PPR is set properly. ● Check whether the brake keeps released during no-load auto-tuning. ● Check whether the inspection button is released before the PMSM with-load auto-tuning is finished.	5A
E20	Speed feedback incorrect	● The encoder model is incorrect. ● The wiring of the encoder is incorect. ● The current remains large at low speed.	● Check the setting of FH-01. ● Check wirin of all signal cables of the encoder. ● Check whether the cable connection of C, D signals of the SIN/COS encoder or cable connection of the U, V, W signals of the UVW encoder is broken. ● Check that there is no mechanical stuck and that the brake has been released during running. ● Check whether the brake is released during running.	5A

Fault Code	Name	Possible Causes	Solution	Level
E22	Leveling signal abnormal	The leveling position deviation is too large in elevator auto-running state.	● Check whether the leveling and door zone sensors work properly. ● Check the installation verticality and depth of the leveling plates. ● Check the leveling signal input points of the SSL-7000-A1. ● Check whether the steel rope slips.	1A
E24	RTC clock fault	The RTC clock information of the SSL-7000-A1 is abnormal.	● Replace the clock battery. ● Replace the SSL-7000-A1.	3B
E25	Storage data abnormal	The storage data of the SSL-7000-A1 is abnormal.	Contact the agent or Monarch.	4A
E26	Earthquake signal	The earthquake signal is active and the duration exceeds 2s.	Check that the earthquake signal is consistent with the parameter setting (NC, NO) of the SSL-7000-A1.	3B
E29	Shorting PMSM stator feedback abnormal	The shorting PMSM stator feedback is abnormal.	● Check that the state (NO, NC) of the feedback contact on the contactor is correct. ● Check that the output indicator on the SSL-7000-A1 acts accordingly when the contactor acts. ● Check that the contactor and corresponding feedback contact act correctly.	5A
E30	Elevator position abnormal	● The elevator automatic running time is too long. ● The elevator re-leveling time is too long. ● The up and down limit switches act during re-leveling. ● The steel rope slips or motor locked-rotor occurs.	● Check whether the up and down limit switches mis-act during re-leveling. ● Check whether the leveling signal cables are connected reliably and whether the signal copper wires may touch the ground or be short circuited with other signal cables. ● Check whether the distance between two floors is too large, causing too long re-leveling running time. ● Check whether the setting of F9-02 is improper (larger than the normal-speed running time). ● Check whether signal loss exists in the encoder circuits.	4A
E31	Emergency running abnormal	The running time in emergency state is too long.	Check whether the emergency power capacity is proper. Check whether the emergency running speed is set properly.	5A
E33	Elevator speed abnormal	The speed exceeds the limit during running.	● Check whether the encoder is used properly. ● Check the setting of motor nameplate parameters. Perform motor auto-tuning again. ● Check the inspection switch and connection of signal cables. ● Check wehther the inspection signal is active during normal speed running.	5A
E34	Logic fault	Logic of the SSL-7000-A1 is abnormal.	Contact the agent or Monarch.	5A

Fault Code	Name	Possible Causes	Solution	Level
E35	Shaft auto-tuning data abnormal	● When shaft auto-tuning is started, the elevator is not at the bottom floor. ● No leveling signal is received within 45s continuous running. ● The floor height is too small. ● The actual number of floors is different from the seting. ● The recoded floor pulses are abnormal. ● The system is not in the inspection state when shaft auto-tuning is performed. ● It is judged upon power-on that shaft auto-tuning is not performed.	1. If E35 is reported when the RUN contactor is not closed, check: ● Whether the next slow-down switch is valid ● Whether F4-01 (Current floor) is set to 1 ● Whether the inspection switch is in inspection state and supports inspection running ● Whether F0-00 is set to 1 2. If E35 is reported at the first leveling position, check: ● Whether the value of F4-03 increases in UP direction and decreases in DOWN direction. If not, exchange the PGA and PGB of the SSL-7000-A1 ● Whether the NO/NC setting of the leveling sensor is set correctly ● Whether the leveling plates are inserted properly if the leveling sensor signal blinks 3. If E35 is reported during running, check: ● Whether the running times out: No leveling signal is received when the running time exceeds F9-02 ● Whether the super short floor function is enabled when the floor distance is less than 50 cm ● Whether the setting of F6-00 (Top floor of the elevator) is smaller than the actual condition 4: If E35 is reported when the elevator arrives at the top floor, check: ● Whether the obtained top floor of the elevator and bottom floor of the elevator are consistent with the setting of F6-00 and F6-01 when the up slow-down signal is valid and the elevator reaches the door zone ● Whether the obtained floor distance is less than 50 cm ● 5. If E35 is reported at power-on, check whether the detected flag length is 0.	4C
E36	RUN contactor feedback abnormal	● The feedback of the RUN contactor is active, but the contactor has no output. ● The controller outputs the RUN signal but receives no RUN feedback. ● When both feedback signals of the RUN contactor are enabled, their states are	● Check whether the feedback contact of the contactor acts properly. ● Check the signal feature (NO, NC) of the feedback contact. ● Check whether the output cables UVW of the controller are connected properly. ● Check whether the control circuit of the RUN contactor coil is normal.	5A

Fault Code	Name	Possible Causes	Solution	Level
		inconsistent.		
E37	Brake contactor feedback abnormal	● The output of the brake contactor is inconsistent with the feedback. ● When both feedback signals of the brake contactor are enabled, their states are inconsistent.	● Check whether the brake coil and feedback contact are correct. ● Check the signal feature (NO, NC) of the feedback contact. ● Check whether the control circuit of the brake contactor coil is normal.	5A
E38	Encoder signal abnormal	● There is no input of the encoder pulses when the elevator runs automatically. ● The direction of the input encoder signal is incorrect when the elevator runs automatically. ● F0-00 is set to 0 (SVC) in distance control.	● Check whether the encoder is used correctly. ● Exchange phases A and B of the encoder. ● Check the setting of F0-00, and change it to CLVC. ● Check whether the system and signal cables are grounded reliably. ● Check whether cabling between the encoder and the PG card is correct.	5A
E39	Motor overheat	The motor overheat relay input remains valid for a certain time.	● Check whether the thermal protection relay is normal. ● Check whether the motor is used properly and whether it is damaged. ● Improve cooling conditions of the motor.	3A
E40	Elevator running reached	The set elevator running time is reached.	Check the related parameter, or contact the agent or Monarch.	4B
E41	Safety circuit disconnected	The safety circuit signal becomes OFF.	● Check the safety circuit switches and their states. ● Check whether the external power supply is normal. ● Check whether the safety circuit contactor acts properly. ● Confirm the signal feature (NO, NC) of the feedback contact of the safety circuit contactor.	5A
E42	Door lock disconnected during running	The door lock circuit feedback is invalid during the elevator running.	● Check whether the hall door lock and the car door lock are in good contact. ● Check whether the door lock contactor acts properly. ● Check the signal feature (NO, NC) of the feedback contact on the door lock contactor. ● Check whether the external power supply is normal.	5A

Fault Code	Name	Possible Causes	Solution	Level
E43	Up limit signal abnormal	The up limit switch acts when the elevator is running in the up direction.	● Check the signal feature (NO, NC) of the up limit switch. ● Check whether the up limit switch is in good contact. ● Check whether the limit switch is installed at a relatively low position and acts even when the elevator arrives at the terminal floor normally.	4C
E44	Down limit signal abnormal	The down limit switch acts when the elevator is running in the down direction.	● Check the signal feature (NO, NC) of the down limit switch. ● Check whether the down limit switch is in good contact. ● Check whether the limit switch is installed at a relatively high position and thus acts even when the elevator arrives at the terminal floor normally.	4C
E45	Slow-down switch position abnormal	● The slow-down signal is abnormal. ● The obtained slow-down distance is incorrect during shaft auto-tuning.	● Check whether the up slow-down 1 and the down slow-down 1 are in good contact. ● Check the signal feature (NO, NC) of the up slow-down 1 and the down slow-down 1. ● Ensure that the obtained slow-down distance satisfies the slow-down requirement at the elevator speed.	4B
E46	Re-leveling abnormal	● The re-leveling running speed exceeds 0.1 m/s. ● The elevator is out of the door zone when re-leveling. ● The feedback of the shorting door lock circuit contactor is abnormal during running.	● Check the original and secondary wiring of the shorting door lock circuit relay. ● Check whether the shorting door lock circuit contactor feedback function is enabled and whether the feedback signal is normal. ● Check whether the encoder is used properly.	2B
E47	Shorting door lock circuit contactor abnormal	● The feedback of the shorting door lock circuit contactor is abnormal when the door pre-open and re-leveling functions are enabled. ● The output of the shorting door lock circuit contactor times out. ● The elevator speed exceeds the limit during running in the condition that the shorting door lock circuit function is enabled.	● Check the signal feature (NO, NC) of the feedback contact on the shorting door lock circuit contactor. ● Check whether the shorting door lock circuit contactor acts properly. ● Check whether the pre-open and re-leveling speeds are set properly.	2B
E48	Door open fault	The consecutive times that the door does not open to the limit reaches the setting in FB-13.	● Check whether the door machine system works properly. ● Check whether the SSL7020 is normal. ● Check whether the door open limit	5A

Fault Code	Name	Possible Causes	Solution	Level
			signal is normal.	
E49	Door close fault	The consecutive times that the door does not open to the limit reaches the setting in FB-13.	● Check whether the door machine system works properly. ● Check whether the SSL7020 is normal. ● Check whether the door lock acts properly.	5A
E51	CAN communication abnormal	Feedback data of CANbus communication with the SSL7020 remains incorrect.	● Check the communication cable connection. ● Check the power supply of the SSL7020. ● Check whether the 24 V power supply of the controller is normal. ● Check whether strong-power interference on communication exists.	1A
E52	SSL7040 communication abnormal	Feedback data of Modbus communication with the SSL7040 remains incorrect.	● Check the communication cable connection. ● Check whether the 24 V power supply of the controller is normal. ● Check whether the SSL7040 addresses are repeated. ● Check whether strong-power interference on communication exists.	1A
E53	Door lock fault	In the automatic running state, the relevant door lock signal is abnormal.	● Check whether the door lock circuit is normal. ● Check whether the feedback contact of the door lock contactor acts properly. ● Check whether the system receives the door open limit signal when the door lock signal is valid. ● Check whether the detected hall door lock state and car door lock state are inconsistent.	5A
E55	Stop at another landing floor	During automatic running of the elevator, the door open limit is not achieved at the present floor.	Check the door open limit signal at the present floor.	1A
E57	Serial peripheral interface (SPI) communication abnormal	The SPI communication is abnormal.	Check the wiring between the control board and the drive board. Contact the agent or Monarch.	5A

Fault Code	Name	Possible Causes	Solution	Level
E58	Shaft position switches abnormal	The up limit and down limit switches are valid simultaneously. The up slow-down and down slow-down switches are valid simultaneously.	● Check whether the states (NO, NC) of the slow-down switches and limit switches are consistent with the parameter setting of the SSL-7000-A1. ● Check whether malfunction of the slow-down switches and limit switches exists.	4B
E60	Leveling signal abnormal	Certain leveling signals are lost.	● Check wiring of the leveling sensor ● Check whether the NO/NC setting of the leveling sensor is set correctly.	3B
E61	Leveling signal abnormal	The leveling signals are all lost.	● Check wiring of the leveling sensor ● Check whether the NO/NC setting of the leveling sensor is set correctly.	3A
E62	Brake feedback abnormal	Left/right brake feedback abnormal	● Check the micro switch in the box of break ● Check whether the gap of brake Is too large or too small ● Check whether control circuit of left/right brake and a detection cable is loose	5A
E63	Lift NO.is not consistent	1.call board No. and FP-05, FP-06 value are different	● Please contact the agent or manufacturer	
E65	UCMP alarm	Car accident protection start	● In the state of repair, can be manually reset ● Check the upper/low level sensor signal is normal This err can be reset in maintenance	
E66	The braking force is unqualified	The braking force is unqualified	● Check the brake force ● Check the friction of the main engine brake ● Check the brake voltage is normal	
E81	Internet communication abnormal	1.Internet module and one machine communication disconnect 5 minutes ; 2.The Internet module disconnected from the remote server for 5 minutes;	● Check the internet interface communication is normal; ● Check the Internet is in normal communication with the computer;	
E82	Internet communication abnormal	Please check the Internet	● Check the internet interface communication is normal; ● Check the Internet is in normal communication with the computer;	

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